# Section_starting_page_1

*Campbell Biology, AP Edition, 12e*

Campbell Biology, twelfth edition cover.

# Campbell Biology

Campbell Biology, twelfth edition cover.

**TWELFTH EDITION**

**Lisa A. Urry**

MILLS COLLEGE AT NORTHEASTERN UNIVERSITY, OAKLAND, CALIFORNIA

**Michael L. Cain**

NEW MEXICO STATE UNIVERSITY

**Steven A. Wasserman**

UNIVERSITY OF CALIFORNIA, SAN DIEGO

**Peter V. Minorsky**

MERCY COLLEGE, DOBBS FERRY, NEW YORK

**Rebecca B. Orr**

COLLIN COLLEGE, PLANO, TEXAS

**Kerry L. Hull**

BISHOP'S UNIVERSITY, QUEBEC, CANADA

Pearson Logo.

Director, Global Higher Ed Content Management and Strategy, Science & Health Sciences: *Jeanne Zalesky*

Manager, Higher Ed Global Content Strategy, Life Sciences: *Joshua Frost*

Associate Content Analyst: *Chelsea Noack*

Director, Higher Ed Product Management, Life Sciences: *Michael Gillespie*

Product Manager: *Rebecca Berardy Schwartz*

Managing Producer: *Michael Early*

Senior Content Producer: *Lori Newman*

Director, Content Development & Partner Relationships: *Ginnie Simione Jutson*

Supervising Editors: *Beth N. Winickoff, Pat Burner*

Senior Developmental Editors: *John Burner, Mary Ann Murray, Hilair Chism, Andrew Recher, Mary Hill*

Developmental Editor, AP Edition: *Jennifer Angel*

Director, Product Management and Marketing, AP, Honors & Electives–Science & Humanities: *Elaine Shema*

Marketing Manager, AP, Honors & Electives–Science & Humanities: *Michele Gillis*

Specialist, Instructional Design and Development: *Sarah Young-Dualan*

Senior Content Developer, Mastering Biology: *Sarah Jensen*

Content Producers, Mastering Biology: *Kaitlin Smith, Ashley Gordon*

Supervising Media Producer: *Tod Regan*

Media Producer: *Ziki Dekel*

Full-Service Vendor: *Integra Software Services, Inc.*

Design Manager:*Mark Ong*

Cover & Interior Designer: *Jeff Puda*

Illustrators: *Lachina Creative*

Rights & Permissions Project Manager: *Matt Perry, SPi Global*

Rights & Permissions Manager:*Ben Ferrini*

Photo Researcher: *Maureen Spuhler*

Manufacturing Buyer: *Stacey Weinberger, LSC Communications*

Cover Photo Credit: *Robert Rohrbaugh/Rohrbaugh Photography*

Copyright © 2023, 2021, 2018, 2014 by Pearson Education, Inc. or its affiliates, 221 River Street, Hoboken, NJ 07030. All Rights Reserved. Manufactured in the United States of America. This publication is protected by copyright, and permission should be obtained from the publisher prior to any prohibited reproduction, storage in a retrieval system, or transmission in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise. For information regarding permissions, request forms, and the appropriate contacts within the Pearson Education Global Rights and Permissions department, please visit www.pearsoned.com/permissions/.

Acknowledgments of third-party content appear in the Credits section, which constitutes an extension of this copyright page.

PEARSON, ALWAYS LEARNING, MasteringTM Biology, and BioFlix® are exclusive trademarks owned by Pearson Education, Inc. or its affiliates in the U.S. and/or other countries.

AP® is a trademark registered and/or owned by the College Board, which was not involved in the production of, and does not endorse, this product.

Unless otherwise indicated herein, any third-party trademarks that may appear in this work are the property of their respective owners and any references to third-party trademarks, logos or other trade dress are for demonstrative or descriptive purposes only. Such references are not intended to imply any sponsorship, endorsement, authorization, or promotion of Pearson's products by the owners of such marks, or any relationship between the owner and Pearson Education, Inc. or its affiliates, authors, licensees or distributors.

**Library of Congress Cataloging-in-Publication Data**

Names: Urry, Lisa A., author. | Cain, Michael L. (Michael Lee), author. | Wasserman, Steven Alexander, author. | Minorsky, Peter V., author. | Orr, Rebecca B., author. | Campbell, Neil A., Biology.

Title: Campbell biology / Lisa A. Urry, Michael L. Cain, Steven A. Wasserman, Peter V. Minorsky, Rebecca B. Orr, Neil A. Campbell.

Description: Twelfth edition. | New York, NY : Pearson, 2020. | Includes index.

Identifiers: LCCN 2019039139 | ISBN 9780135188743 (hardcover) | ISBN 9780135988046 (ebook)

Subjects: LCSH: Biology.

Classification: LCC QH308.2 .C34 2020 | DDC 570—dc23

LC record available at https://lccn.loc.gov/2019039139

**ScoutAutomatedPrintCode**

ISBN 10: 0-13-648687-8 (High School Binding)

ISBN 13: 978-0-13-648687-9 (High School Binding)

www.savvas.com/advanced

Pearson Logo.

---

# Pearson's Commitment to Accessibility

**Accessibility**

We are also committed to providing products that are fully accessible to all learners. As per Pearson's guidelines for accessible educational Web media, we test and retest the capabilities of our products against the highest standards for every release, following the WCAG guidelines in developing new products for copyright year 2022 and beyond.

- You can learn more about Pearson's commitment to accessibility at **https://www.pearson.com/us/accessibility.html**.

**Contact Us**

While we work hard to present unbiased, fully accessible content, we want to hear from you about any concerns or needs with this Pearson product so that we can investigate and address them.

- Please contact us with concerns about any potential bias at **https://www.pearson.com/report-bias.html**.
- For accessibility-related issues, such as using assistive technology with Pearson products, alternative text requests, or accessibility documentation, email the Pearson Disability Support team at **disability.support@pearson.com**.

image

---

# Spotlight on AP® Course Framework Components

Throughout the AP® edition of *Campbell Biology*12e, many features, exercises, and assessments reinforce the AP® Course Framework's four Big Ideas and six Science Practices.

## AP® Course Framework Units

The College Board divides the AP® course content into eight units:

**UNIT 1:** Chemistry of Life

**UNIT 2:** Cell Structure and Function

**UNIT 3:** Cellular Energetics

**UNIT 4:** Cell Communication and Cell Cycle

**UNIT 5:** Heredity

**UNIT 6:**Gene Expression and Regulation

**UNIT 7:** Natural Selection

**UNIT 8:** Ecology

## Big Ideas

Four Big Ideas build conceptual understanding and spiral throughout the eight units:

**BIG IDEA 1: EVOLUTION (EVO)**

The process of evolution drives the diversity and unity of life.

**BIG IDEA 2: ENERGETICS (ENE)**

Biological systems use energy and molecular building blocks to grow, reproduce, and maintain dynamic homeostasis.

**BIG IDEA 3: INFORMATION STORAGE AND TRANSMISSION (IST)**

Living systems store, retrieve, transmit, and respond to information essential to life processes.

**BIG IDEA 4: SYSTEMS INTERACTIONS (SYI)**

Biological systems interact, and these systems and their interactions exhibit complex properties.

Each unit is further broken down into enduring understandings and learning objectives. Beneath the learning objectives, statements of essential knowledge identify what students should know and be able to do. All the AP test questions derive from the learning objectives.

## Science Practices

Six Science Practices essential to the study of biology can be combined with *any* learning objective in an AP test question:

**SCIENCE PRACTICE 1**

***Concept Explanation***

Explain biological concepts, processes, and models presented in written format.

**SCIENCE PRACTICE 2**

***Visual Representations***

Analyze visual representations of biological concepts and processes.

**SCIENCE PRACTICE 3**

***Questions and Methods***

Determine scientific questions and methods.

**SCIENCE PRACTICE 4**

***Representing and Describing Data***

Represent and describe data.

**SCIENCE PRACTICE 5**

***Statistical Tests and Data Analysis***

Perform statistical tests and mathematical calculations to analyze and interpret data.

**SCIENCE PRACTICE 6**

***Argumentation***

Develop and justify scientific arguments using evidence.

# Correlation to AP® Course Framework

The College Board's Advanced Placement® Biology Course Framework (effective Fall 2019) organizes content into eight units of study grounded in four Big Ideas: Evolution, Energetics, Information Storage and Transmission, and Systems Interactions. The chart below lists the Key Concepts in *Campbell Biology* 12e, AP® Edition, that teach the Essential Knowledge contributing to the Enduring Understandings for each of the Framework's eight units. All the details of the Framework, plus a full correlation to specific page numbers of this textbook, can be found at PearsonSchool.com/AdvancedCorrelations.

A table lists the items under the AP Course Framework Units 1 through 3.

| **A P, registered trademark, Course Framework Unit 1: Chemistry of Life** |
|---|
| A P, registered trademark, Big Ideas |
| Energetics (E N E) |
| Information Storage and Transmission (I S T) |
| Systems Interactions (S Y I) |
| A P, registered trademark, Course Framework Unit 2: Cell Structure and Function |
| A P, registered trademark, Big Ideas |
| Evolution (E V O) |
| Energetics (E N E) |
| E N E-2: Cells have membranes that allow them to establish and maintain internal environments that are different from their external environments. |
| Systems Interactions (S Y I) |
| A P, registered trademark, Course Framework Unit 3 Cellular Energetics |
| A P, registered trademark, Big Ideas |
| Energetics (E N E) |
| Systems Interactions (S Y I) |

A table lists the items under the AP Course Framework Units 4 through 7.

| A P, registered trademark, Course Framework Unit 4: Cell |
| --- |
| A P, registered trademark, Big Ideas |
| Energetics (E N E) |
| Information Storage and Transmission (I S T) |
| I S T-3: Cells communicate by generating, transmitting, receiving, and responding to chemical signals. |
| A P, registered trademark, Course Framework Unit 5 Heredity |
| A P, registered trademark, Big Ideas |
| Evolution (E V O) |
| Information Storage and Transmission (I S T) |
| Systems Interactions (S Y I) |
| A P, registered trademark, Course Framework Unit 6 Gene Expression and Regulation |
| A P, registered trademark, Big Ideas |
| Information Storage and Transmission (I S T) |
| I S T-2: Differences in the expression of genes account for some of the phenotypic differences between organisms. |
| I S T-4: The processing of genetic information is imperfect and is a source of genetic variation. |
| A P, registered trademark, Course Framework Unit 7: Natural Selection |
| A P, registered trademark, Big Ideas |
| Evolution (E V O) |
| E V O-2: Organisms are linked by lines of descent from common ancestry. |
| E V O-3: Life continues to evolve within a changing environment. |
| Systems and Interactions (S Y I) |

A table lists the items under the AP Course Framework Unit 8: Ecology.

| A P, registered trademark, Course Framework Unit 8: Ecology |
| --- |

| |
|---|
| A P, registered trademark, Big Ideas |
| Evolution (E V O) |
| Energetics (E N E) |
| E N E-3: Timing and coordination of biological mechanisms involved in growth, reproduction, and homeostasis depend on organisms responding to environmental cues. |
| E N E-4: Communities and ecosystems change on the basis of interactions among populations and disruptions to the environment. |
| Information Storage and Transmission (I S T) |
| Systems Interactions (S Y I) |
| S Y 1-2: Competition and cooperation are important aspects of biological systems. |
| S Y 1-3: Naturally occurring diversity among and between components within biological systems affects interactions with the environment. |

# About the Authors

The author team's contributions reflect their biological expertise as researchers and their teaching sensibilities gained from years of experience as instructors at diverse institutions. They are also experienced textbook authors, having written *Campbell Biology in Focus* in addition to *Campbell Biology*.

A group photo of the authors. From left to right, Rebecca Orr, Lisa A. Urry, Michael Cain, Steven Wasserman, and Peter Minorsky.

A photograph of Lisa A. Urry, one of the authors of Campbell biology.

**Lisa A. Urry** (Chapter 1 and Units 1–3) is Professor of Biology and Co-Chair of the Biology Department at Mills College at Northeastern University in Oakland, California. After earning a B.A. at Tufts University, she completed her Ph.D. at the Massachusetts Institute of Technology (MIT). Lisa has conducted research on gene expression during embryonic and larval development in sea urchins. Deeply committed to promoting opportunities in science for women and underrepresented minorities, she has taught courses ranging from introductory and developmental biology to an immersive course on the U.S./Mexico border.

A photograph of Michael Cain, one of the authors of Campbell biology.

**Michael L. Cain** (Units 4, 5, and 8) is an ecologist and evolutionary biologist who is now writing full-time. Michael earned an A.B. from Bowdoin College, an M.Sc. from Brown University, and a Ph.D. from Cornell University. As a faculty member at New Mexico State University, he taught introductory biology, ecology, evolution, botany, and conservation biology. Michael is the author of dozens of scientific papers on topics that include foraging behavior in insects and plants, long-distance seed dispersal, and speciation in crickets. He is also a coauthor of an ecology textbook.

A photograph of Steven Wasserman, one of the authors of Campbell biology.

**Steven A. Wasserman** (Unit 7) is Professor of Biology at the University of California, San Diego (UCSD). He earned an A.B. from Harvard University and a Ph.D. from MIT. Working on the fruit fly *Drosophila*, Steve has done research on developmental biology, reproduction, and immunity. Having taught genetics, development, and physiology to undergraduate, graduate, and medical students, he now focuses on introductory biology, for which he has been honored with UCSD's Distinguished Teaching Award.

A photograph of Peter Minorsky, one of the authors of Campbell biology.

**Peter V. Minorsky** (Unit 6) is Professor of Biology at Mercy College in New York, where he teaches introductory biology, ecology, and botany. He received his A.B. from Vassar College and his Ph.D. from Cornell University. Peter taught at Kenyon College, Union College, Western Connecticut State University, and Vassar College; he is also the science writer for the journal *Plant Physiology*. His research interests concern how plants sense environmental change. Peter received the 2008 Award for Teaching Excellence at Mercy College.

A photograph of Rebecca Orr, one of the authors of Campbell biology.

**Rebecca B. Orr** (Ready-to-Go Teaching Modules, Interactive Visual Activities, eTextbook Media Integration) is Professor of Biology at Collin College in Plano, Texas, where she teaches introductory biology. She earned her B.S. from Texas A&M University and her Ph.D. from University of Texas Southwestern Medical Center at Dallas. Rebecca has a passion for investigating strategies that result in more effective learning and retention, and she is a certified Team-Based Learning Collaborative Trainer Consultant. She enjoys focusing on the creation of learning opportunities that both engage and challenge students.

A photograph of Kerry L. Hull, one of the authors of Campbell biology.

**Kerry L. Hull** (Unit 7) is the Dean of Science and a Professor in the Department of Biology and Biochemistry at Bishop's University in Quebec, Canada. She holds a B.Sc. in Biology and a Ph.D. in Physiology (Endocrinology), both from the University of Alberta (Canada). She teaches Physiology and Anatomy courses to undergraduates, using guided inquiry and case studies as her primary teaching approaches. A comparative endocrinologist by training, her current research focuses on the development and implementation of active learning strategies in Anatomy and Physiology courses.

Neil Campbell.

**Neil A. Campbell** (1946–2004) earned his M.A. from the University of California, Los Angeles, and his Ph.D. from the University of California, Riverside. His research focused on desert and coastal plants. Neil's 30 years of teaching included introductory biology courses at Cornell University, Pomona College, and San Bernardino Valley College, where he received the college's first Outstanding Professor Award in 1986. For many years he was also a visiting scholar at UC Riverside. Neil was the founding author of *Campbell Biology*.

> ***In memory of Pat Burner, editor extraordinaire, honorary co-author, and dear friend, whose dedication to the quality ofCampbell Biologywas unparalleled.***

**LAU, MLC, SAW, PVM, RBO, and KLH**

---

# Preface

Decorative image.

We are honored to present the Twelfth Edition of *Campbell Biology*. For the last three decades, *Campbell Biology* has been the leading college text in the biological sciences. It has been translated into 19 languages and has provided millions of students with a solid foundation in college-level biology. This success is a testament not only to Neil Campbell's original vision but also to the dedication of hundreds of reviewers who, together with editors, artists, and contributors, have shaped and inspired this work.

## New to This Digital Update

The Digital Update to the eTextbook of *Campbell Biology*, Twelfth Edition includes current, relevant content updates and engaging interactive resources.

- **New Content:** New material throughout the eTextbook includes coverage of:
- SARS-CoV-2 and the COVID-19 pandemic, including a new figure on mRNA vaccines, as well as expanded discussions of vaccine campaigns, vaccine hesitancy, and emerging and zoonotic pathogens (see Chapters 19, 43, and 54)
- updated data on global climate and population changes (see Chapters 3, 55, and 56)

- updates in gene editing, stem cell research, genomics, and cancer genetics (see Chapters 17, 18, and 20)
- **CheckPoint Questions:** Embedded at point-of-use in the eTextbook, these short sequences of interactive questions diagnose and remediate misconceptions around topics that are common "stumbling blocks" to understanding. Customized feedback guides students through an evidence-based approach to reason their way to learning and retaining the content. The CheckPoints are also assignable in Mastering Biology.
- **Interactive Visual Activities:** These activities guide students in navigating through complex figures as they interact with the art, view embedded video clips, and gauge their understanding with self-check questions. These are embedded in the Pearson eTextbook for seamless access, and instructors can assign them with assessment questions through Mastering Biology. The Interactive Visual Activities include the following:
- **Figure 6.32** Visualizing the Scale of the Molecular Machinery of the Cell
- **Figure 13.8** Exploring Meiosis in an Animal Cell
- **Figure 23.18** Make Connections: The Sickle-Cell Allele
- **Figure 40.5** Exploring Structure and Function in Animal Tissues
- **Figure 55.19** Make Connections: The Working Ecosystem

## New to the Twelfth Edition

Our goals for the Twelfth Edition include:

- **supporting students** with new visual presentations of content and new study tools
- **supporting instructors** by providing new teaching modules with tools and materials for introducing, teaching, and assessing important and often challenging topics
- **integrating text and media** to engage, guide, and inform students in an active process of inquiry and learning

Here we provide an overview of the new features that we have developed for the Twelfth Edition.

- **NEW! Chapter Openers Re-envisioned.** Catalyzed by feedback from students and instructors, informed by data analytics, and building on the results of science education research, we have redesigned the opening of every chapter of the text. The result is more visual, more

interactive, and more engaging. In place of an opening narrative, the first page of each chapter is organized around three new elements that provide students with the specific tools and approaches needed to achieve the learning objectives of that chapter:

- **NEW! Visual Overviews.** Centered on a basic biological question related to the opening photo and legend, the Visual Overview illustrates a core idea of the chapter with straightforward art and text. Students get an immediate sense of what the chapter is about and what kinds of thinking will underlie its exploration.
- **NEW! Study Tips.** Just as the Visual Overview introduces students to *what* they will learn, the study tip offers guidance in *how* to learn. It encourages students to learn actively through such proven strategies as drawing a flow chart, labeling a diagram, or making a table. Each tip provides an effective strategy for tackling important content in the chapter.
- **NEW! Highlights of Digital Resources.** In conversations with users of the textbook, we often encounter a limited awareness of the digital tools the text provides to facilitate instruction and learning. We therefore created *Go to Mastering Biology*, a chapter opener section where we highlight some of the tutorials, animations, and other interactives available for students to exploreon their own or for instructors to assign. These resources include Get Ready for This Chapter questions, Figure Walkthroughs, HHMI BioInteractive videos, Ready-to-Go Teaching Modules, and more.
- **NEW! Updated Content.** As in each new edition of*Campbell Biology*, the Twelfth Edition incorporates new content, summarized in Highlights of New Content. Content updates reflect rapid, ongoing changes in knowledge about climate change, genomics, gene-editing technology (CRISPR), evolutionary biology, microbiome-based therapies, emerging viruses, and more.
- **NEW! Active Reading Guides.** These worksheets provide students with self-assessment activities to complete as they read each chapter. Students can download the Active Reading Guides from the Mastering Biology Study Area.
- **5 NEW! Ready-to-Go Teaching Modules.**The Readyto- Go Teaching Modules provide instructors with active learning exercises and questions to use in class, plus Mastering Biology assignments that can be assigned before and after class. A total of 15 modules are now available in the Instructor Resources area of Mastering Biology.

## Pearson eTextbook

Students using the Pearson eTextbook will reap all the benefits of the new text features, while also benefiting from the following new and existing interactive resources, which are integrated directly into the online text:

- **NEW!** An expanded collection of the popular **Figure Walkthroughs** guide students through key figures with narrated explanations and figure mark-ups that reinforce important points.
- **EXPANDED! 500 animations and videos** bring biology to life. These include new resources from **HHMI BioInteractive** that engage students in topics from CRISPR to coral reefs.
- **Get Ready for This Chapter** interactive questions provide a quick check of student understanding of the background information needed to learn a new chapter's content, with feedback to bolster their preparation.
- **Vocabulary Self-Quizzes** and **Practice Tests** at the end of each chapter provide opportunities for students to test their understanding.
- Links to **Interviews** from all editions of *Campbell Biology* are included in the chapter where they are most relevant. The interviews show students the human side of science by featuring diverse scientists talking about how they became interested in biology and what inspires them.

## Mastering Biology

**Mastering Biology** provides valuable resources for instructors to assign homework and for students to study on their own:

- **Assignments.** Mastering Biology is the most widely used online assessment and tutorial program for biology, providing an extensive library of thousands of tutorials and questions that are graded automatically.
- **AP NEW AND IMPROVED! Pearson LabBench**features 13 investigations that align closely with the AP® Biology Course Framework and reinforce key concepts from lab outside the laboratory.
- **AP Exam-style Practice Questions**test student mastery of the Framework.
- Hundreds of self-paced **tutorials** provide individualized coaching with specific hints and feedback on the most difficult topics in the course.
- Optional **Adaptive Follow-up Assignments** provide additional questions tailored to each student's needs.
- **Pearson eTextbook**. The Pearson eTextbook, described above, can be directly accessed from Mastering Biology.

- **Dynamic Study Modules.** These popular review tools can be assigned, or students can use them for self-study.
- **Study Area.** Media references in the printed book direct students to the wealth of online self-study resources available to them in the Mastering Biology Study Area, including Active Reading Guides, Figure Walkthroughs, videos, animations, Get Ready for This Chapter, Practice Tests, Cumulative Test, and more.
- **Instructor Resources.**This area of Mastering Biology provides one-stop shopping for Ready-to-Go Teaching Modules, PowerPoints, Clicker Questions, animations, videos, the Test Bank, AP® Exam-style Questions, and more.

For more information, visit www.masteringbiology.com.

## Our Hallmark Features

Teachers of general biology face a daunting challenge: to help students acquire a conceptual framework for organizing an ever-expanding amount of information. The hallmark features of *Campbell Biology* provide such a framework, while promoting a deeper understanding of biology and the process of science. As such, they are well-aligned with the core competencies outlined by the **Vision and Change** national conferences. Furthermore, the core concepts defined by Vision and Change have close parallels in the unifying themes that are introduced in Chapter 1 and integrated throughout the book.

Chief among the themes of both Vision and Change and *Campbell Biology* is **evolution.** Each chapter of this text includes at least one Evolution section that explicitly focuses on evolutionary aspects of the chapter material, and each chapter ends with an Evolution Connection Question and a Write About a Theme Question.

To help students distinguish “the forest from the trees,” each chapter is organized around a framework of three to seven carefully chosen **Key Concepts**. The text, Concept Check Questions, Summary of Key Concepts, and Mastering Biology resources all reinforce these main ideas and essential facts.

Because text and illustrations are equally important for learning biology, **integration of text and figures** has been a hallmark of *Campbell Biology* since the First Edition. The new Visual Overviews, together with our popular Visualizing Figures, Exploring Figures, and Make Connections Figures, epitomize this approach.

To encourage **active reading** of the text, *Campbell Biology* includes numerous opportunities for students to stop and think about what they are reading, often by putting pencil to paper to draw a sketch, annotate a

figure, or graph data. Answering these questions requires students to write or draw as well as think and thus helps develop the core competency of communicating science.

Finally, *Campbell Biology* has always featured **scientific inquiry**. The inquiry activities provide students practice in applying the process of science and using quantitative reasoning, addressing core competencies from Vision and Change.

## Our Partnership with Instructors and Students

The real test of any textbook is how well it helps instructors teach and students learn. We welcome comments from both students and instructors. Please address your suggestions to:


- Lisa Urry (Chapter 1 and Units 1–3): lurry@mills.edu
- Michael Cain (Units 4, 5, and 8): mlcain@nmsu.edu
- Peter Minorsky (Unit 6): pminorsky@mercy.edu
- Steven Wasserman (Unit 7): stevenw@ucsd.edu
- Rebecca Orr (Media): rorr@collin.edu


---

# Highlights of New Content

This section highlights selected new content in *Campbell Biology*, Twelfth Edition. In addition to the content updates noted here, every chapter has a **new Visual Overview** on the chapter opening page.

## Unit 1 The Chemistry of Life

In Unit 1, new content engages students in learning foundational chemistry. Chapter 2 includes a new micrograph of the tiny hairs on a gecko's foot that allow it to walk up a wall. The opening photo for Chapter 3 features a ringed seal, a species endangered by the melting of Arctic sea ice due to climate change. Chapter 3 also has added coverage on the discovery of a large subsurface reservoir of liquid water on Mars and the first CO2 enhancement study done on an unconfined natural coral reef (both reported in 2018). Chapter 4 now includes the discovery of carbon-based compounds on Mars reported by NASA in 2018. In Chapter 5, the technique of cryo-electron microscopy is introduced, due to its increasing importance in the determination of molecular structure.

# Unit 2 The Cell

Our main goal for this unit was to make the material more accessible, inviting, and exciting to students. Chapter 6 includes a new text description of cryo-electron microscopy (cryo-EM) and a new cryo-EM image in Figure 6.3. Art has been added to Figure 6.17 to illustrate the dynamic nature of mitochondrial networks. Chapter 7 begins with a new chapter-opening image showing neurotransmitter release during exocytosis. **Figure 8.1** includes a new photo of bioluminescent click beetle larvae on the outside of a termite mound and a new Visual Overview that illustrates how the laws of thermodynamics apply to metabolic reactions like bioluminescence.

**Figure 8.1**

Figure 8.1 shows a diagram of thermodynamics and the biological process.

Chapter 9 includes new information on human brown fat usage, the role of fermentation during the production of chocolate, and recent research on the role of lactate in mammalian metabolism. Chapter 10 begins with a new concept that puts photosynthesis into a big-picture ecological context. Chapter 10 also includes a discussion of the 2018 discovery of a new form of chlorophyll found in cyanobacteria that can carry out photosynthesis using far-red light. In Chapter 11, the relevance of synaptic signaling is underscored by mentioning that it is a target for treatment of depression, anxiety, and PTSD. In Chapter 12, the cell cycle figure (Figure 12.6) now includes cell images and labels describing the events of each phase.

# Unit 3 Genetics

Chapters 13–17 incorporate changes that help students grasp the more abstract concepts of genetics and their chromosomal and molecular underpinnings. For example, a new Concept Check 13.2 question asks students about shoes as an analogy for chromosomes. In Chapter 14, the classic idea of a single gene determining hair or eye color, or even earlobe attachment, is discussed as an oversimplification. Also, the “Fetal Testing” section has been updated to reflect current practices in obstetrics. Chapter 15 now includes new information on “three-parent” babies. In Concept 16.3, the text and Figure 16.23 have been extensively revised to reflect recent models of the structure and organization of interphase chromatin, as well as how chromosomes condense during preparation for mitosis. Chapter 17 now describes the mutation responsible for the albino phenotype of the Asinara donkeys featured in the chapter-opening photo. To make it easier to cover CRISPR, a new section has been added to Concept 17.5 describing the CRISPR-Cas9 system, including Figure 17.28, “Gene editing using the CRISPR-Cas9 system” (formerly Figure 20.14).

Chapters 18–21 are extensively updated, driven by exciting new discoveries based on DNA sequencing and gene-editing technology. In Chapter 18, the coverage of epigenetic inheritance has been enhanced and updated. Also in Chapter 18, a description of topologically associated domains has been added, along with an update on the 4D Nucleome Network. In Chapter 19, the topic of emerging viral diseases has been updated extensively to discuss SARS-CoV-2 and the COVID-19 pandemic, as well as to clearly differentiate influenza viruses that are emerging from those that cause seasonal flu. Other Chapter 19 updates include information on vaccine programs and the correlation between vaccine hesitancy and viral outbreaks. In addition, the discussion of the emergence and spread of HIV has been updated to discuss its origin in sub-Saharan Africa and subsequent transmission to other regions, including the social context for its worldwide spread. Chapter 20 has been extensively updated, including addition of two new subsections, “Personal Genome Analysis” and “Personalized Medicine,” with new information on direct-to-consumer genome analysis. Other updates include the first cloning of a primate, stem cell treatment of age-related macular degeneration, CRISPR correction of the sickle-cell disease allele in mice, and a report of gene editing of fertilized human eggs that resulted in live births. Chapter 21 updates include results of the Cancer Genome Atlas Project, a newly discovered function of retrotransposon transcription, and new information on the *FOXP2* gene.

## Unit 4 Mechanisms of Evolution

The revision of Unit 4 uses an evidence-based approach to strengthen how we help students understand key evolutionary concepts. For example, new text in Concept 24.3 describes how hybrids can become reproductively isolated from both parent species, leading to the formation of a new species. Evidence supporting this new material comes from a 2018 study on the descendants of hybrids between two species of Galápagos finches and provides an example of how scientists can observe the formation of a new species in nature. In Concept 25.2, the discussion of fossils as a form of scientific evidence is supported by a new figure (Figure 25.5) that highlights five different types of fossils and how they are formed. The unit also features new material that connects evolutionary concepts and societal issues. For example, in Chapter 23, new text and a new figure **(Figure 23.19)** describe how some snowshoe hare populations have not adapted to ongoing climate change, causing them to be poorly camouflaged in early winter and leading to increased mortality. Additional changes include a new section of text in Chapter 22 and a figure (Figure 22.22) describing biogeographical evidence for evolution in a group of freshwater fishes that cannot survive in salt water, yet live in regions separated by wide stretches of ocean. In Chapter 25, a new figure (Figure 25.11) provides fossil evidence of an enormous change in the evolutionary history of life: the first appearance of large, multicellular eukaryotes.

**Figure 23.19**

```
Lack of variation in a population can limit adaptation.
```

Figure 23.19 shows two photos of two different rabbits.

## Unit 5 The Evolutionary History of Biological Diversity

In keeping with our goal of developing students' skills in interpreting visual representations in biology, we have added a new Visualizing Figure, Figure 32.8, "Visualizing Animal Body Symmetry and Axes." New Visual Skills Questions provide practice on topics such as interpreting phylogenetic trees and using graphs to infer how rapidly antibiotic resistance evolves in bacteria. Chapter 31 has been significantly revised to account for new fossil discoveries and updates to the phylogenetic tree of fungi (Figure 31.10). Chapter 34 has been updated with recent genomic data and fossil discoveries indicating that Neanderthals and Denisovans are more closely related to each other than to humans and that they interbred with each other (and with humans), including two new figures (Figures 34.51 and 34.52b). In Chapter 29, a new figure (Figure 29.1) provides a visual overview of major steps in the colonization of land by plants, and revisions to text in Concept 29.1 strengthen our description of derived traits of plants that facilitated life on land. Chapter 27 includes a new section of text that describes the rise of antibiotic resistance and multidrug resistance and discusses novel approaches in the search for new antibiotics. This new material is supported by two new figures, **Figure 27.22** and Figure 27.23. Other updates include the revision of many phylogenies to reflect recent phylogenomic data; a new Inquiry Figure (Figure 28.26) on the root of the eukaryotic tree; and new text describing the 2017 discovery of 315,000-year-old fossils of a hominin that had facial features like those of humans, while the back of its skull was elongated, as in earlier species.

**Figure 27.22**

```
The rise of antibiotic resistance.
```

Figure 27.22 shows a horizontal stacked bar graph of antibiotic resistance.

The x-axis (years): 1930 to 2010 in increments of 20 units. The y-axis: Antibiotics: Ceftaroline, Daptomycin, Linezolid, Imipenem, Vancomycin, Methicillin, Colistin, Erythromycin, Tetracycline, and Penicillin. Each stacked bar contains antibiotic discovered (grey circle), medical usage begins (black circle), and resistance observed (red bar). Ceftaroline: antibiotic discovered (2001), medical usage begins (2010), and resistance observed (after 2010). Daptomycin: antibiotic discovered (1988), medical usage begins (2001), and resistance observed (2005). Linezolid: antibiotic discovered (1998), medical usage begins (2000), and resistance observed (after 2001). Imipenem: antibiotic discovered (1980), medical usage begins (1985), and

resistance observed (1988). Vancomycin: antibiotic discovered (1953), medical usage begins (1972), and resistance observed (1989). Methicillin: antibiotic discovered (1958), medical usage begins (1959), and resistance observed (1962). Colistin: antibiotic discovered (1949), medical usage begins (1959), and resistance observed (2011). Erythromycin: antibiotic discovered (1948), medical usage begins (1952), and resistance observed (1968). Tetracycline: antibiotic discovered (1942), medical usage begins (1950), and resistance observed (1959). Penicillin: antibiotic discovered (1928), medical usage begins (1942), and resistance observed (1940).

## Unit 6 Plant Form and Function

In Chapter 35, greater emphasis is placed on how structure fits function in vascular plants by way of a new Visual Overview. In Chapter 36, a new Visual Skills Question provides a quantitative exercise in estimating stomatal density. Chapter 37 begins with an emphasis on the importance of crop fertilization in feeding the world. To increase student engagement, renewed emphasis is placed on the link between the nutrition of plants and the nutrition of the organisms, including humans, that feed on them. Table 37.1 concerning plant essential elements has been expanded to include micronutrients as well as macronutrients. In Concept 37.2, a new subsection titled “Global Climate Change and Food Quality” discusses new evidence that global climate change may be negatively impacting the nutritional mineral content of crops. In Chapter 38, the discussion of genetic engineering and agriculture has been enhanced by a discussion of biofortification and by updates concerning “Golden Rice.” Chapter 39 includes new updates on the location of the IAA receptor in plant cells and the role of abscisic acid in bud dormancy. The introduction to Concept 39.2 has been revised to emphasize that plants use many classes of chemicals in addition to the classic hormones to communicate information.

## Unit 7 Animal Form and Function

The Unit 7 revisions feature pedagogical innovations coupled with updates for currency. A striking new underwater image of Emperor penguins (Figure 40.1) opens the unit and highlights the contributions of form, function, and behavior to homeostasis in general as well as to the specific topic of thermoregulation. The artwork used to introduce and explore homeostasis throughout the unit (Figures 40.8, 40.17, 41.23, 42.28, 44.19, 44.21, and 45.18) has been improved and refined to provide a clear and consistent presentation of the role of perturbation in triggering a response. The Unit 7 revision adds coverage in Chapter 43 and elsewhere of how SARS-CoV-2 infection affects the human body and a new figure (Figure 43.23X) showing how mRNA vaccines interact with the immune system. Also in Chapter 43, the introduction of the adaptive immune response has been shifted to later in the chapter, allowing students to build on the features of innate

immunity before tackling the more demanding topic of the adaptive response. In Chapter 48, the structural overview of neurons is now completed before the introduction of information processing. A new illustration, **Figure 49.8**, provides a concise visual comparison of sympathetic and parasympathetic neurons with each other and with motor neurons of the CNS. In addition, in-depth consideration of glia is now provided in Concept 49.1, where it is more logically integrated into the overview of nervous systems. At the end of the unit, an eye-catching photograph of the male frigatebird's courtship display (Figure 51.1) introduces the topic of animal behavior. Among the content updates that enhance currency and student engagement throughout the unit are discussions of phage therapy and fecal transplantation, state-of-the-art treatments that both rely on microbiome data, and chronic traumatic encephalopathy (CTE), as well as the latest findings on dinosaur locomotion (Concept 40.1), the awarding of a Nobel Prize in 2017 in the field of circadian rhythms (Concept 40.2), and reference to the ongoing public health crisis of opioid addiction in the context of considering the brain's reward system (Concept 49.5).

**Figure 49.8**

```
    Comparison of pathways in the motor and autonomic nervous
systems.
```

Figure 49.8 shows two diagrams of the motor and autonomic nervous systems.

## Unit 8 Ecology

Complementary goals of the Unit 8 revision were to strengthen our coverage of core concepts while also increasing our coverage of how human actions affect ecological communities. Revisions include a new section of text and a new figure (Figure 52.7) on how plants (and deforestation) can affect the local or regional climate; a new section of text in Concept 55.1 that summarizes how ecosystems work; new text and a new figure (Figure 52.25) illustrating how rapid evolution can cause rapid ecological change; new material in Concept 55.2 on eutrophication and how it can cause the formation of large "dead zones" in aquatic ecosystems; and new text and a new figure (Figure 54.22) on how the abundance of organisms at each trophic level can be controlled by bottom-up or top-down control. A new figure **(Figure 56.23)** shows the extent of the record-breaking 2017 dead zone in the Gulf of Mexico and the watershed that contributes to its nutrient load. Chapter 54 includes updates on emerging diseases and the COVID-19 pandemic, Concept 56.1 includes a new section that describes attempts to use cloning to resurrect species lost to extinction, and Concept 56.4 includes a new section of text and two new figures (Figure 56.27 and 56.28) on plastic waste, a major and growing environmental problem. In keeping with our book-wide goal of expanding our coverage of climate change, Chapter 56 has a new Scientific Skills Exercise in which students interpret changes in

atmospheric $CO_2$ concentrations. Chapter 55 describes how climate warming is causing large regions of tundra in Alaska to release more $CO_2$ than they absorb (thereby contributing to further climate warming); a new figure (Figure 56.32) describes human and natural factors that contribute to rising global temperatures; and a new section of text in Concept 56.4 describes how global climate change models are developed and why they are valuable.

**Figure 56.23**

```
    A dead zone arising from nitrogen pollution in the
Mississippi basin.
```

Figure 56.23 shows two maps of nitrogen pollution.

---

# Comprehensive Student and Instructor Resources

## Mastering™ Biology Access

Mastering™ Biology, the most widely used online biology tutorial and homework program, provides an extensive library of activities, tutorials, and homework assignments. Mastering™ Biology with Pearson eText uses the Socratic method to coach students through problem-solving techniques, offering hints and simpler questions on request to help students learn, not just practice. Tutorials guide students through the most challenging biology topics and help them make connections. Upon textbook purchase, students and teachers are granted access to Mastering™ Biology with Pearson eText. High school teachers can obtain preview or adoption access to Mastering™ Biology in one of the following ways:

### Preview Access

- Teachers can request preview access online by visiting www.PearsonSchool.com/Access_Request. Select Science, choose Initial Access, and complete the form under Option 2. Preview Access information will be sent to the teacher via e-mail.

### Adoption Access

- With the purchase of this program, a Pearson Adoption Access Card with Instructor Manual will be delivered with your textbook purchase. (ISBN: 978-0-13-354087-1)
- Ask your sales representative for a Pearson Adoption Access Card with Instructor Manual. (ISBN: 978-0-13-354087-1)

OR

- Visit PearsonSchool.com/Access_Request, select Science, choose initial Access, and complete the form under Option 3—MyLab/ Mastering Class Adoption Access. Teacher and Student access information will be sent to the teacher via e-mail.

### Students, ask your teacher for access

Pearson reserves the right to change and/or update technology platforms, including possible edition updates to customers during the term of access. This will allow Pearson to continue to deliver the most up-to-date content and technology to customers. Customers will be notified of any change prior to the beginning of the new school year.

## Student Resources

The following resources are available for purchase.

### AP NEW! Pearson's Test Prep Series for AP®****Biology

by Fred and Theresa Holtzclaw, The Webb School of Knoxville 978-0-134-54690-2/0-134-54690-3

This Test Prep Workbook for AP® Biology is designed to support the AP® Biology Course Framework. Each chapter reviews content that is Essential Knowledge and makes connections to the Framework. Material that may be used as Illustrative Examples is included and clearly distinguished from the required content. This workbook will help students become familiar with the Big Ideas, apply the Science Practices, gain insight into the AP® labs, and prepare for the AP® Exam with practice tests. In addition to its student friendly features—You Must Know boxes, Tip From the Readers boxes, and practice problems—the workbook now integrates Science Practices throughout and includes an emphasis on making connections between topics.

### NEW! Active Reading Guide, Twelfth Edition

by Fred and Theresa Holtzclaw, The Webb School of Knoxville

Designed to be used side-by-side with the Student Edition, the Pearson Active Reading Guide helps students tackle some of the toughest aspects of an AP® Biology course—reading for understanding, active learning, and mastering vocabulary—through self-assessment activities that students complete as they read each chapter. This resource is available within Mastering™ Biology, which is accessible with the adoption of this program.

## Inquiry in Action: Interpreting Scientific Papers, Fourth Edition*

by Ruth V. Buskirk, University of Texas at Austin, and Christopher M. Gillen, Kenyon College

978-0-134-47861-6/0-134-47861-4

This guide helps students learn how to read and understand primary research articles. Part A presents complete articles accompanied by questions that help students analyze the article. Related Inquiry Figures are included in the supplement. Part B covers every part of a research paper, explaining the aim of the sections and how the paper works as a whole.

## Practicing Biology: A Student Workbook, Sixth Edition*

by Jean Heitz and Cynthia Giffen, University of Wisconsin, Madison

978-0-134-48603-1/0-134-48603-X

This workbook offers a variety of activities to suit different learning styles. Activities such as modeling and concept mapping allow students to visualize and understand biological processes. Other activities focus on basic skills, such as reading and drawing graphs.

## Biological Inquiry: A Workbook of Investigative Cases, Fifth Edition*

by Margaret Waterman, Southeast Missouri State University, and Ethel Stanley, BioQUEST Curriculum Consortium and Beloit College

978-0-134-48646-8/0-134-48646-3

This workbook offers ten investigative cases. Each case study requires students to synthesize information from multiple chapters of the text and apply that knowledge to a real-world scenario as they pose hypotheses, gather new information, analyze evidence, graph data, and draw conclusions. A link to a student website is in the Study Area in Mastering Biology.

### Study Guide, Eleventh Edition

by Martha R. Taylor, Ithaca, New York, and Michael Pollock, Mount Royal University

978-0-134-44377-5/0-134-44377-2

This popular study aid provides concept maps, chapter summaries, word roots, and a variety of interactive activities including multiple choice, short-answer essay, art labeling, and graph interpretation questions.

### Study Card, Eleventh Edition

978-0-134-48648-2/0-134-48648-X

This quick-reference card provides students with an overview of the entire field of biology, helping them see the connections between topics.

### Spanish Glossary for Biology

By Laura P. Zanello, University of California, Riverside

978-0-32183498-0/0-321-83498-4

For all the glossary terms, this resource provides definitions in Spanish.

## Instructor Resources

### The Instructor Resources Area of Mastering Biology

- **5 NEW! Ready-to-Go Teaching Modules** help instructors efficiently make use of the available teaching tools for the toughest topics. Before-class assignments, in-class activities, and after-class assignments are provided for ease of use. Instructors can incorporate active learning into their course with the suggested activity ideas and clicker questions or Learning Catalytics Questions. A total of 15 modules are now available.
- **Lecture Presentations in PowerPoint®** for each chapter with lecture notes, editable figures (art and photos with enlarged, customizable labels), tables, and links to animations and videos:

Concept 7.1 Cellular membranes are fluid mosaics of lipids and proteins.

- Phospholipids are the most abundant lipid in most membranes.
- Phospholipids are amphipathic molecules, containing hydrophobic and hydrophilic regions.

- A phospholipid bilayer can exist as a stable boundary between two aqueous compartments.

Decorative image.

Cross section of the phospholipid bilayer.

Phospholipid bilayer with hydrophilic heads drawn as circles and two squiggly hydrophobic tails attached to each head. Tails point towards each other so that only the heads contact water. Cutaway diagram zooms in on the structure of two molecules in the phospholipid bilayer, where hydrophilic heads interact with water molecules on both sides of the membrane, but the hydrophobic tails are tucked away from water.

The structure of an idealized flower.

At the top of the stem is the receptacle. On this is the ovary, in which are the ovules. Sepals extend to either side of the receptacle. Around the ovary are petals, and within this are stamen and single carpel (simple pistil). Stamen has a filament and an anther. Single carpel includes the ovary, the style from the ovary, and the stigma at the end of the style.

3 D structure of a protein.

Do fungal endophytes benefit a woody plant?

Experiment. Fungal endophytes are symbiotic fungi found within the bodies of all plants examined to date. Researchers tested whether fungal endophytes benefit the cacao tree (Theobroma cacao). This tree, whose name means food of the gods in Greek, is the course of the beans used to make chocolate, and it is cultivated throughout the tropics. A particular mixture of fungal endophytes was added to the leaves of some cacao seedlings, but not others. (In cacao, fungal endophytes colonize leaves after the seedling germinates.) The seedlings were then inoculated with a virulent pathogen, the protist Phytophthora.

Results. Fewer leaves were killed by the pathogen in seedlings with fungal endophytes than in seedlings without endophytes. Among leaves that survived, pathogens damaged less of the leaf surface area in seedlings with endophytes than in seedlings without endophytes. Two bar charts showing leaf mortality (%) and Leaf area damaged (%) is twice as high for endophytes not present, pathogen present versus both endophyte and pathogen present. Leaf mortality (%) for Endophytes not present, pathogen present (E minus P plus) is roughly 25 versus roughly 10 for both endophyte and pathogen present (E plus P plus). Leaf area damaged (%) for E minus P plus is roughly 15 versus roughly 7.5 for E plus P+ plus.

Conclusion. The presence of endophytes appears to benefit cacao trees by reducing the leaf mortality and damage caused by Phytophthora.

- **Accessible Lecture Presentations in PowerPoint** with alt text for every image; students can access alt text with a screen reader if needed

- **Editable Images in PowerPoint** (all art and photos from the text) and all tables from the text in PowerPoint with enlarged, customizable labels
- **Labeled and Unlabeled JPEG Images,** including art, photos from the text, and extra photos
- **Clicker Questions in PowerPoint,** which can be used to stimulate effective classroom discussions (for use with or without clickers):

A multiple choice question on the intermediate disturbance hypothesis.

New Zealand ecologists recorded the numbers of invertebrate species in streambeds that experienced different intensities and frequencies of flooding. The largest numbers of species were present in streambeds with blank disturbances.

A. no

B. mild

C. intermediate

D. intense

Graph of number of taxa versus index of disturbance intensity. The x-axis is index of disturbance intensity (log scale), ranging from 0.9 to 2.0. The y-axis is number of taxa, ranging from 10 to 35. The curve is hump-shaped, with peak number of taxa at intermediate disturbance levels.

A multiple choice question on evapotranspiration.

Evapotranspiration, the evaporation of water from soil and plants, is much higher in hot areas with abundant rainfall than in cooler or low precipitation areas. Which statement describes the relationship between potential evapotranspiration and vertebrate species richness?

A. As evapotranspiration increases, the number of vertebrate species declines.

B. Abundant rainfall in hot areas has an undetectable effect on the number of vertebrate species.

C. As evapotranspiration increases, so does vertebrate species richness.

Based on the data, one expects fewer vertebrate species in areas of high temperature and abundant rainfall than in areas with lower temperature and less rainfall.

Graph of vertebrate species richness versus potential evapotranspiration. The x-axis is potential evapotranspiration in millimeters per year, ranging from 0 to 2000 with increments every 500. The y-axis is vertebrate species richness (log scale), ranging from 10 to 200. Vertebrate species richness increases with increasing potential evapotranspiration, from 10 species near 0 millimeters per year to 100 species at 1000

millimeters per year, after which point both vertebrate species richness and potential evapotranspiration begin to level off. At both 1500 and 2000 millimeters per year, there are about 150 vertebrate species.

- **EXPANDED! 500 Instructor Animations and Videos,** including BioFlix 3-D Animations, HHMI BioInteractive Animations and Videos, BBC Videos, and much more
- **Test Bank** questions in TestGen® software and Microsoft® Word. This extensively revised resource contains over 4,500 questions, including scenario-based questions and art, graph, and data interpretation questions. **NEW!** Every image in the Test Bank has alt text, which students can access with a screen reader if needed.
- **AP** In addition to hundreds of test bank questions, Mastering offers **AP®Exam-style Practice Questions** that test student mastery of the Framework.
- **NEW! Statistics Worksheets for Biology**
- **Instructor Answers** to Scientific Skills Exercises, Problem-Solving Exercises, Interpret the Data Questions, and Essay Questions; includes rubric and tips for grading short-answer essays
- **Instructor Guides for Supplements:** Instructor Guide for *Active Reading Guides*; Instructor Guide for *Practicing Biology: A Student Workbook*; Instructor Guide for *Biological Inquiry: A Workbook of Investigative Cases*; Answer Key for *Inquiry in Action: Interpreting Scientific Papers; Investigating Biology Lab Prep Guide*; and *Investigating Biology* Lab Data Tables

## Learning CatalyticsTM

Learning Catalytics allows students to use their smartphone, tablet, or laptop to respond to questions in class. For more information, visit learningcatalytics.com.

Decorative image.

## Learning Management Systems

Integration with various learning management systems is available for Mastering Biology. Contact your sales representative for details.

An Instructor Guide is available for downloading through the Instructor Resources area of Mastering Biology.

An Instructor Guide is available for downloading through the Instructor Resources area of Mastering Biology.

An Instructor Guide is available for downloading through the Instructor Resources area of Mastering Biology.

# Brief Contents


- ****1 Evolution, the Themes of Biology, and Scientific Inquiry
- ****Unit 1 The Chemistry of Life

---

- ****2 The Chemical Context of Life
- ****3 Water and Life
- ****4 Carbon and the Molecular Diversity of Life
- ****5 The Structure and Function of Large Biological Molecules
- ****Unit 2 The Cell
- ****6 A Tour of the Cell
- ****7 Membrane Structure and Function
- ****8 An Introduction to Metabolism
- ****9 Cellular Respiration and Fermentation
- ****10 Photosynthesis
- ****11 Cell Communication
- ****12 The Cell Cycle
- ****Unit 3 Genetics
- ****13 Meiosis and Sexual Life Cycles
- ****14 Mendel and the Gene Idea
- ****15 The Chromosomal Basis of Inheritance
- ****16 The Molecular Basis of Inheritance
- ****17 Gene Expression: From Gene to Protein
- ****18 Regulation of Gene Expression

- ****19** Viruses
- ****20** DNA Tools and Biotechnology
- ****21** Genomes and Their Evolution
- ****Unit 4** Mechanisms of Evolution
- ****22** Descent with Modification: A Darwinian View of Life
- ****23** The Evolution of Populations
- ****24** The Origin of Species
- ****25** The History of Life on Earth
- ****Unit 5** The Evolutionary History of Biological Diversity
- 
    - ****26** Phylogeny and the Tree of Life
- ****27** Bacteria and Archaea
- ****28** Protists
- ****29** Plant Diversity I: How Plants Colonized Land
- ****30** Plant Diversity II: The Evolution of Seed Plants
- ****31** Fungi
- ****32** An Overview of Animal Diversity
- ****33** An Introduction to Invertebrates
- ****34** The Origin and Evolution of Vertebrates
- ****Unit 6** Plant Form and Function
- ****35** Vascular Plant Structure, Growth, and Development
- ****36** Resource Acquisition and Transport in Vascular Plants
- ****37** Soil and Plant Nutrition
- ****38** Angiosperm Reproduction and Biotechnology
- ****39** Plant Responses to Internal and External Signals
- ****Unit 7** Animal Form and Function

- ****40 Basic Principles of Animal Form and Function
- ****41 Animal Nutrition
- ****42 Circulation and Gas Exchange
- ****43 The Immune System
- ****44 Osmoregulation and Excretion
- ****45 Hormones and the Endocrine System
- ****46 Animal Reproduction
- ****47 Animal Development
- ****48 Neurons, Synapses, and Signaling
- ****49 Nervous Systems
- ****50 Sensory and Motor Mechanisms
- ****51 Animal Behavior
- ****Unit 8 Ecology
- ****52 An Introduction to Ecology and the Biosphere
- ****53 Population Ecology
- ****54 Community Ecology
- ****55 Ecosystems and Restoration Ecology
- ****56 Conservation Biology and Global Change


---

# Acknowledgments

The authors wish to express their gratitude to the global community of instructors, researchers, students, and publishing professionals who have contributed to the Twelfth Edition of *Campbell Biology*.

As authors of this text, we are mindful of the daunting challenge of keeping up to date in all areas of our rapidly expanding subject. We are grateful to the many scientists who helped shape this text by discussing their research fields with us, answering specific questions in their areas of expertise, and sharing their ideas

about biology education. We are especially grateful to the following, listed alphabetically: Graham Alexander, Elizabeth Atkinson, Kristian Axelsen, Ron Bassar, Christopher Benz, David Booth, George Brooks, Abby Dernberg, Jean DeSaix, Alex Engel, Rachel Kramer Green, Fred Holtzclaw, Theresa Holtzclaw, Tim James, Kathy Jones, Azarias Karamanlidis, Gary Karpen, Joe Montoya, Laurie Nemzer, Kevin Peterson, T. K. Reddy, David Reznick, Thomas Schneider, Alastair Simpson, Martin Smith, Steven Swoap, and John Taylor. In addition, the biologists listed in the Reviewers section provided detailed reviews, helping us ensure the text's scientific accuracy and improve its pedagogical effectiveness. Thanks also to Mary Camuso and Ann Sinclair for contributing a creative Study Tip for their fellow students.

Thanks also to the other professors and students, from all over the world, who contacted the authors directly with useful suggestions. We alone bear the responsibility for any errors that remain, but the dedication of our consultants, reviewers, and other correspondents makes us confident in the accuracy and effectiveness of this text.

Interviews with prominent scientists have been a hallmark of *Campbell Biology* since its inception, and conducting these interviews was again one of the great pleasures of revising the book. To open the eight units of this edition, we are proud to include interviews with Kenneth Olden, Diana Bautista, Francisco Mojica, Cassandra Extavour, Penny Chisholm, Dennis Gonsalves, Steffanie Strathdee, and Chelsea Rochman.

Mastering Biology and the other electronic accompaniments for this text are invaluable teaching and learning aids. We are grateful to the contributors for the Ready-to-Go Teaching Modules: Chad Brassil, Ruth Buskirk, Eileen Gregory, Angela Hodgson, Molly Jacobs, Bridgette Kirkpatrick, Maureen Leupold, Jennifer Metzler, Karen Resendes, Justin Shaffer, Allison Silveus, Jered Studinski, Cynthia Surmacz, Sara Tallarovic, and Carole Twichell. We would also like to extend our sincere appreciation to Carolyn Wetzel for her hard work on the Figure Walkthroughs. And our gratitude goes to Bryan Jennings and Roberta Batorsky for their work on the Reading Questions. Thanks also to Ann Brokaw and Bob Cooper for their contributions to the AAAS Science in the Classroom activities; we also appreciate the support of Beth Reudi, Shelby Lake, and Lydia Kaprelian from AAAS.

The value of *Campbell Biology* as a learning tool is greatly enhanced by the supplementary materials that have been created for instructors and students. We recognize that the dedicated authors of these materials are essentially writing mini (and not so mini) books. We appreciate the hard work and creativity of all the authors listed, with their creations, in the Student and Lab Supplements section. We are also grateful to Kathleen Fitzpatrick and Nicole Tunbridge (PowerPoint® Lecture Presentations); Roberta Batorsky, Douglas Darnowski, James Langeland, and David Knochel (Clicker Questions); Sonish Azam, Ford Lux, Karen Bernd, Janet Lanza, Chris Romero, Marshall Sundberg, Justin Shaffer, Ed Zalisko, and David Knochel (Test Bank).

*Campbell Biology* results from an unusually strong synergy between a team of scientists and a team of publishing professionals.

Our editorial team at Pearson Education again demonstrated unmatched talents, commitment, and pedagogical insights.

Josh Frost, our Manager of Higher Ed Global Content Strategy for Life Sciences, brought publishing savvy, intelligence, and a much-appreciated level head to leading the whole team. The clarity and effectiveness of every page owe much to our extraordinary Supervising Editors Beth Winickoff and Pat Burner, who worked with a top-notch team of Senior Developmental Editors in John Burner, Mary Ann Murray, Hilair Chism, Andrew Recher, and Mary Hill. Our unsurpassed Director of Content Development Ginnie Simione Jutson and Project Manager Rebecca Berardy Schwartz were indispensable in moving the project in the right direction. We also want to thank Robin Heyden for organizing the annual Biology Leadership Conferences and keeping us in touch with the world of AP Biology, and Beth Wilbur for her many years of wise editorial leadership. We also extend our thanks to Nida Ahmed, Associate Content Strategy Administrator.

You would not have this beautiful text if not for the work of the production team: Director, Content Production & Digital Studio Erin Gregg; Managing Producer Michael Early; Senior Content Producer Lori Newman; Photo Researcher Maureen Spuhler; Copy Editor Joanna Dinsmore; Proofreader Pete Shanks; Rights & Permissions Manager Ben Ferrini; Rights & Permissions Project Manager Matt Perry; the staff at Integra Software Services, Inc., and Lachina Creative; Design Manager Mark Ong; Text and Cover Designer Jeff Puda; and Manufacturing Buyer Stacey Weinberger. We also thank those who worked on the text's supplements: Project Manager Shiny Rajesh and her team at Integra Software Services.

For creating the wonderful package of electronic media that accompanies the text, we are grateful to Senior Content Developer Sarah Jensen; Content Producers Kaitlin Smith and Ashley Gordon; Director, Production & Digital Studio Katie Foley; Director, Production & Digital Studio Laura Tommasi; Supervising Media Producer Tod Regan; Specialist, Instructional Design and Development Sarah Young-Dualan; Digital Program Manager, Science, Caroline Ayres; Project Manager Katie Cook; Media Producer Ziki Dekel; Manager, Creative Technology Greg Davis; and Senior Learning Tools Strategist Kassi Foley.

For their important roles in marketing the text and media, we thank Sierra Stewart, Kate Gittins, Alysun Estes, Kelly Galli, Jane Campbell, Brad Parkins, and Stacey Abraham. For their enthusiasm, encouragement, and support, we are grateful to Michael Gillespie, Director, Higher Ed Product Management, Life Sciences and Adam Jaworski, VP Product Management Higher Ed.

The Pearson sales team, which represents *Campbell Biology* on campus, is an essential link to the users of the text. They tell us what you like and don't like about the text, communicate the features of the text, and provide prompt service. We thank them for their hard work and professionalism. For representing our text

to our international audience, we thank our sales and marketing partners throughout the world. They are all strong allies in biology education.

Finally, we wish to thank our families and friends for their encouragement and patience throughout this long project. Our special thanks to Lily and Alex (L.A.U.); Debra and Hannah (M.L.C.); Aaron, Sophie, Noah, and Gabriele (S.A.W.); Natalie (P.V.M.); Jim, Abby, Dan, and Emily (R.B.O.), and Norman, Lauren, and Evan (K.L.H.). Thanks to Jane Reece, now retired, for her generosity and thoughtfulness throughout her many years as a Campbell author. And, as always, thanks to Rochelle, Allison, Jason, McKay, and Gus.

Lisa A. Urry, Michael L. Cain, Steven A. Wasserman,

Peter V. Minorsky, Rebecca B. Orr, and Kerry L. Hull

# Reviewers

## AP® Advisory Group

Many expert reviewers contributed to this textbook. In addition, an Advisory Group of experienced teachers provided valuable advice for the AP® Edition:

Jean DeSaix, *University of North Carolina, Chapel Hill*

Fred Holtzclaw, *The Webb School of Knoxville*

Theresa Holtzclaw, *The Webb School of Knoxville*

Mike Judge, *Sacred Heart Schools*

Nancy Monson, *West Linn High School*

Nancy Ramos, *Northside Health Careers High School*

Peggy O'Neill Skinner, *The Bush School*

Diane Sweeney, *Sacred Heart Schools*

Brad Williamson, *University of Kansas (emeritus)*

## Twelfth Edition Reviewers

Sheena Abernathy, *College of the Mainland*; James Arnone, *William Paterson University*; Josh Auld, *West Chester University*; Gemma Bartha, *Springfield College*; Louise Beard, *University of Essex*; Marin Beaupre, *Massasoit Community College*; Kevin Bennett, *University of Hawai'i*; Kelsie Bernot, *North Carolina A&T State University*; Christine Bezotte, *Elmira College*; Chris Bloch, *Bridgewater State University*; Aiwei Borengasser, *University of Arkansas – Pulaski Tech*; Robert Borgon, *University of Central Florida*; Nicole Bournias-Vardiabasis, *California State University, San Bernardino*; George Brooks, *University of California, Berkeley*; Michael Buoni, *Delaware Technical Community College*; Kelcey Burris, *Union High School*; Elena Cainas, *Broward College*; Mickael Cariveau, *Mount Olive College*; Billy Carver, *Lees-McRae College*; Anne Casper, *Eastern Michigan University*; Bruce Chase, *University of Nebraska, Omaha*; Amanda Chau, *Blinn College*; Katie Clark, *University of California, Riverside*; Catharina Coenen, *Allegheny College*; Curt Coffman, *Vincennes University*; Juliet Collins, *University of Wisconsin, Madison*; Bob Cooper, *Pennsbury High School*; Robin Cotter, *Phoenix College*; Marilyn Cruz-Alvarez, *Florida Gulf Coast University*; Noelle Cutter, *Molloy College*; Deborah Dardis, *Southeastern Louisiana University*; Farahad Dastoor, *University of Maine, Orono*; Andrew David, *Clarkson University*; Jeremiah Davie, *D'Youville College*; Brian Deis, *University of Hawai'i* Jean DeSaix, *University of North Carolina at Chapel Hill* Kelly Dubois, *Calvin College*; Cynthia Eayre, *Fresno City College*; Arri Eisen, *Emory University*; Lisa Elfring, *University of Arizona*; Kurt Elliott, *Northwest Vista College*; Ana Esther Escandon, *Los Angeles Harbor College*; Linda Fergusson-Kolmes, *Portland Community College*; April Fong, *Portland Community College*; Robert Fowler, *San Jose State University*; Brittany Gasper, *Florida Southern College*; Carri Gerber, *The Ohio State University*; Marina Gerson, *California State University, Stanislaus*; Brian Gibbens, *University of Minnesota*; Sara Gremillion, *Georgia Southern University*; Ron Gross, *Community College of Allegheny County*; Melissa Gutierrez, *University of Southern Mississippi*; Gokhan Hacisalihoglu, *Florida A&M University*; Monica Hall-Woods, *St. Charles Community College*; Catherine Hartkorn, *New Mexico State University*; Valerie Haywood, *Case Western Reserve University*; Maryann Herman, *St. John Fisher College*; Alexander Heyl, *Adelphi University*; Laura Hill, *University of Vermont*; Anne-Marie Hoskinson, *South Dakota State University*; Katriina Ilves, *Pace University*; James Jacob, *Tompkins Cortland Community College*; Darrel James, *Fred C. Beyer High School*; Jerry Johnson, *Corban University*; Greg Jones, *Santa Fe College*; Kathryn Jones, *Howard Community College*; Seth Jones, *University of Kentucky*; Steven Karafit, *University of Central Arkansas*; Lori Kayes, *Oregon State University*; Ben Kolber, *Duquesne University*; Catherine Konopka, *John Carroll University*; Bill Kroll, *Loyola University, Chicago*; MaryLynne LaMantia, *Golden West College*; Neil Lamb, *HudsonAlpha Institute for Biotechnology*; Michelle LaPorte, *St. Louis Community College*; Neil Lax, *Duquesne University*; John Lepri, *Carrboro High School*; Jani Lewis, *State University of New York at Geneseo*; Eddie Lunsford, *Southwestern Community College*; Alyssa MacDonald, *Leeward Community College*; Charles Mallery, *University of Miami*; Marlee Marsh, *Columbia College*; Nicole McDaniels, *Herkimer College*; Mike Meighan, *University of California, Berkeley*; Jennifer Metzler, *Ball State University*; Grace Ju Miller, *Indiana Wesleyan University*; Terry Miller, *Central Carolina*

*Community College*; Shamone Mizenmayer, *Central High School*; Cam Muir, *University of Hawai'i at Hilo*; Heather Murdock, *San Francisco State University*; Madhavan Narayanan, *Mercy College*; Jennifer Nauen, *University of Delaware*; Karen Neal, *J. Sargeant Reynolds Community College, Richmond*; Leonore Neary, *Joliet Junior College*; Shanna Nifoussi, *University of Superior*; Jennifer Ortiz, *North Hills School District*; Fernanda Oyarzun, *Universidad Católica de la Santísima Concepción, Chile*; Stephanie Pandolfi, *Wayne State University*; John Plunket, *Horry-Georgetown Technical College*; Elena Pravosudova, *University of Nevada, Reno*; Pushpa Ramakrishna, *Chandler-Gilbert Community College*; Sami Raut, *University of Alabama, Birmingham*; Robert Reavis, *Glendale Community College*; Linda Rehfuss, *Bucks County Community College*; Deborah Rhoden, *Snead State Community College*; Linda Richardson, *Blinn College*; Brian Ring, *Valdosta State University*; Rob Ruliffson, *Minneapolis Community and Technical College*; Judy Schoonmaker, *Colorado School of Mines*; David Schultz, *University of Louisville*; David Schwartz, *Houston Community College*; Duane Sears, *University of California, Santa Barbara*; J. Michael Sellers, *University of Southern Mississippi*; Pramila Sen, *Houston Community College*; Jyotsna Sharma, *University of Texas San Antonio*; Joan Sharp, *Simon Fraser University*; Marcia Shofner, *University of Maryland, College Park*; Linda Sigismondi, *University of Rio Grande*; Davida Smyth, *Mercy College*; Helen Snodgrass, *YES Prep North Forest*; Ayodotun Sodipe, *Texas Southern University*; Kathy Sparace, *Tri-County Technical College*; Patricia Steinke, *San Jacinto College Central*; Elizabeth Sudduth, *Georgia Gwinnett College*; Aaron Sullivan, *Houghton College*; Yvonne Sun, *University of Dayton*; Andrea Swei, *San Francisco State University*; Greg Thurmon, *Central Methodist University*; Stephanie Toering-Peters, *Wartburg College*; Monica Togna, *Drexel University*; Gail Tompkins, *Wake Technical Community College*; Tara Turley-Stoulig, *Southeastern Louisiana University*; Bishnu Twanabasu, *Weatherford College*; Erin Ventresca, *Albright College*; Wei Wan, *Texas A&M University*; Alan Wasmoen, *Metropolitan Community College, Nebraska*; Fred Wasserman, *Boston University*; Vicki Watson, *University of Montana*; Bill Wesley, *Mars Area High School*; Clay White, *Lone Star College*; Lisa Whitenack, *Allegheny College*; Larry Wimmers, *Towson University*; Heather Woodson, *Gaston College*; Shelly Wu, *Texas Christian University*; Mary Wuerth, *Tamalpais High School*; John Yoder, *University of Alabama*; Alyson Zeamer, *University of Texas San Antonio*.

---

# Detailed Contents

## 1 Evolution, the Themes of Biology, and Scientific Inquiry


- **CONCEPT 1.1 The study of life reveals unifying themes**

- - Theme: New Properties Emerge at Successive Levels of Biological Organization
- Theme: Life's Processes Involve the Expression and Transmission of Genetic Information
- Theme: Life Requires the Transfer and Transformation of Energy and Matter
- Theme: From Molecules to Ecosystems, Interactions Are Important in Biological Systems
- **CONCEPT 1.2 The Core Theme: Evolution accounts for the unity and diversity of life**
- - Classifying the Diversity of Life
- Charles Darwin and the Theory of Natural Selection
- The Tree of Life
- **CONCEPT 1.3 In studying nature, scientists form and test hypotheses**
- - Exploration and Observation
- Gathering and Analyzing Data
- Forming and Testing Hypotheses
- The Flexibility of the Scientific Process
- *A Case Study in Scientific Inquiry:* Investigating Coat Coloration in Mouse Populations
- Variables and Controls in Experiments
- Theories in Science
- **CONCEPT 1.4 Science benefits from a cooperative approach and diverse viewpoints**
- - Building on the Work of Others
- Science, Technology, and Society

# Unit 1 The Chemistry of Life

**Interview: Kenneth Olden**

# 2 The Chemical Context of Life

- **CONCEPT 2.1 Matter consists of chemical elements in pure form and in combinations called compounds**
- - Elements and Compounds

- The Elements of Life
- *Case Study:*Evolution of Tolerance to Toxic Elements
- **CONCEPT 2.2 An element's properties depend on the structure of its atoms**
- ◦ Subatomic Particles

- Atomic Number and Atomic Mass
- Isotopes
- The Energy Levels of Electrons
- Electron Distribution and Chemical Properties
- Electron Orbitals
- **CONCEPT 2.3 The formation and function of molecules and ionic compounds depend on chemical bonding between atoms**
- ◦ Covalent Bonds

- Ionic Bonds
- Weak Chemical Interactions
- Molecular Shape and Function
- **CONCEPT 2.4 Chemical reactions make and break chemical bonds**

# 3 Water and Life

- **CONCEPT 3.1 Polar covalent bonds in water molecules result in hydrogen bonding**
- **CONCEPT 3.2 Four emergent properties of water contribute to Earth's suitability for life**
- ◦ Cohesion of Water Molecules

- Moderation of Temperature by Water
- Floating of Ice on Liquid Water
- Water: The Solvent of Life
- Possible Evolution of Life on Other Planets
- **CONCEPT 3.3 Acidic and basic conditions affect living organisms**
- ◦ Acids and Bases

- The pH Scale

- Buffers
- Acidification: A Threat to Our Oceans

# 4 Carbon and the Molecular Diversity of Life

- **CONCEPT 4.1 Organic chemistry is key to the origin of life**
- **CONCEPT 4.2 Carbon atoms can form diverse molecules by bonding to four other atoms**
- ◦ The Formation of Bonds with Carbon
- Molecular Diversity Arising from Variation in Carbon Skeletons
- **CONCEPT 4.3 A few chemical groups are key to molecular function**
- ◦ The Chemical Groups Most Important in the Processes of Life
- ATP: An Important Source of Energy for Cellular Processes
- The Chemical Elements of Life: *A Review*

# 5 The Structure and Function of Large Biological Molecules

- **CONCEPT 5.1 Macromolecules are polymers, built from monomers**
- ◦ The Synthesis and Breakdown of Polymers
- The Diversity of Polymers
- **CONCEPT 5.2 Carbohydrates serve as fuel and building material**
- ◦ Sugars
- Polysaccharides
- **CONCEPT 5.3 Lipids are a diverse group of hydrophobic molecules**
- ◦ Fats
- Phospholipids
- Steroids
- **CONCEPT 5.4 Proteins include a diversity of structures, resulting in a wide range of functions**

- - Amino Acids (Monomers)
- Polypeptides (Amino Acid Polymers)
- Protein Structure and Function
- **CONCEPT 5.5 Nucleic acids store, transmit, and help express hereditary information**
- - The Roles of Nucleic Acids
- The Components of Nucleic Acids
- Nucleotide Polymers
- The Structures of DNA and RNA Molecules
- **CONCEPT 5.6 Genomics and proteomics have transformed biological inquiry and applications**
- - DNA and Proteins as Tape Measures of Evolution

# Unit 2 The Cell

**Interview: Diana Bautista**

# 6 A Tour of the Cell

- **CONCEPT 6.1 Biologists use microscopes and biochemistry to study cells**
- - Microscopy
- Cell Fractionation
- **CONCEPT 6.2 Eukaryotic cells have internal membranes that compartmentalize their functions**
- - Comparing Prokaryotic and Eukaryotic Cells
- A Panoramic View of the Eukaryotic Cell
- **CONCEPT 6.3 The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes**
- - The Nucleus: Information Central
- Ribosomes: Protein Factories

- **CONCEPT 6.4 The endomembrane system regulates protein traffic and performs metabolic functions**
- The Endoplasmic Reticulum: Biosynthetic Factory
- The Golgi Apparatus: Shipping and Receiving Center
- Lysosomes: Digestive Compartments
- Vacuoles: Diverse Maintenance Compartments
- The Endomembrane System: *A Review*
- **CONCEPT 6.5 Mitochondria and chloroplasts change energy from one form to another**
- The Evolutionary Origins of Mitochondria and Chloroplasts
- Mitochondria: Chemical Energy Conversion
- Chloroplasts: Capture of Light Energy
- Peroxisomes: Oxidation
- **CONCEPT 6.6 The cytoskeleton is a network of fibers that organizes structures and activities in the cell**
- Roles of the Cytoskeleton: Support and Motility
- Components of the Cytoskeleton
- **CONCEPT 6.7 Extracellular components and connections between cells help coordinate cellular activities**
- Cell Walls of Plants
- The Extracellular Matrix (ECM) of Animal Cells
- Cell Junctions
- **CONCEPT 6.8 A cell is greater than the sum of its parts**


## 7 Membrane Structure and Function


- **CONCEPT 7.1 Cellular membranes are fluid mosaics of lipids and proteins**
- The Fluidity of Membranes
- Evolution of Differences in Membrane Lipid Composition

- Membrane Proteins and Their Functions
- The Role of Membrane Carbohydrates in Cell-Cell Recognition
- Synthesis and Sidedness of Membranes
- **CONCEPT 7.2 Membrane structure results in selective permeability**
- ◦ The Permeability of the Lipid Bilayer
- Transport Proteins
- **CONCEPT 7.3 Passive transport is diffusion of a substance across a membrane with no energy investment**
- ◦ Effects of Osmosis on Water Balance
- Facilitated Diffusion: Passive Transport Aided by Proteins
- **CONCEPT 7.4 Active transport uses energy to move solutes against their gradients**
- ◦ The Need for Energy in Active Transport
- How Ion Pumps Maintain Membrane Potential
- Cotransport: Coupled Transport by a Membrane Protein
- **CONCEPT 7.5 Bulk transport across the plasma membrane occurs by exocytosis and endocytosis**
- ◦ Exocytosis
- Endocytosis

# 8 An Introduction to Metabolism

- **CONCEPT 8.1 An organism's metabolism transforms matter and energy**
- ◦ Metabolic Pathways
- Forms of Energy
- The Laws of Energy Transformation
- **CONCEPT 8.2 The free-energy change of a reaction tells us whether or not the reaction occurs spontaneously**
- ◦ Free-Energy Change, $\Delta G$

- Free Energy, Stability, and Equilibrium
- Free Energy and Metabolism
- **CONCEPT 8.3 ATP powers cellular work by coupling exergonic reactions to endergonic reactions**
- ◦ The Structure and Hydrolysis of ATP
- How ATP Provides Energy That Performs Work
- The Regeneration of ATP
- **CONCEPT 8.4 Enzymes speed up metabolic reactions by lowering energy barriers**
- ◦ The Activation Energy Barrier
- How Enzymes Speed Up Reactions
- Substrate Specificity of Enzymes
- Catalysis in the Enzyme's Active Site
- Effects of Local Conditions on Enzyme Activity
- The Evolution of Enzymes
- **CONCEPT 8.5 Regulation of enzyme activity helps control metabolism**
- ◦ Allosteric Regulation of Enzymes
- Localization of Enzymes Within the Cell


## 9 Cellular Respiration and Fermentation


- **CONCEPT 9.1 Catabolic pathways yield energy by oxidizing organic fuels**
- ◦ Catabolic Pathways and Production of ATP
- Redox Reactions: Oxidation and Reduction
- The Stages of Cellular Respiration: *A Preview*
- **CONCEPT 9.2 Glycolysis harvests chemical energy by oxidizing glucose to pyruvate**
- **CONCEPT 9.3 After pyruvate is oxidized, the citric acid cycle completes the energy-yielding oxidation of organic molecules**
- ◦ Oxidation of Pyruvate to Acetyl CoA
- The Citric Acid Cycle

- **CONCEPT 9.4 During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis**
- ◦ The Pathway of Electron Transport
- Chemiosmosis: The Energy-Coupling Mechanism
- An Accounting of ATP Production by Cellular Respiration
- **CONCEPT 9.5 Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen**
- ◦ Types of Fermentation
- Comparing Fermentation with Anaerobic and Aerobic Respiration
- The Evolutionary Significance of Glycolysis
- **CONCEPT 9.6 Glycolysis and the citric acid cycle connect to many other metabolic pathways**
- ◦ The Versatility of Catabolism
- Biosynthesis (Anabolic Pathways)
- Regulation of Cellular Respiration via Feedback Mechanisms

## 10 Photosynthesis

- **CONCEPT 10.1 Photosynthesis feeds the biosphere**
- **CONCEPT 10.2 Photosynthesis converts light energy to the chemical energy of food**
- ◦ Chloroplasts: The Sites of Photosynthesis in Plants
- Tracking Atoms Through Photosynthesis
- The Two Stages of Photosynthesis: *A Preview*
- **CONCEPT 10.3 The light reactions convert solar energy to the chemical energy of ATP and NADPH**
- ◦ The Nature of Sunlight
- Photosynthetic Pigments: The Light Receptors
- Excitation of Chlorophyll by Light
- A Photosystem: A Reaction-Center Complex Associated with Light-Harvesting Complexes

- Linear Electron Flow
- Cyclic Electron Flow
- A Comparison of Chemiosmosis in Chloroplasts and Mitochondria
- **CONCEPT 10.4 The Calvin cycle uses the chemical energy of ATP and NADPH to reduce** $CO_2$ **to sugar**
- **CONCEPT 10.5 Alternative mechanisms of carbon fixation have evolved in hot, arid climates**
- ◦ Photorespiration: An Evolutionary Relic?
- Plants
- CAM Plants
- **CONCEPT 10.6 Photosynthesis is essential for life on Earth: a review**

# 11 Cell Communication

- **CONCEPT 11.1 External signals are converted to responses within the cell**
- ◦ Evolution of Cell Signaling
- Local and Long-Distance Signaling
- The Three Stages of Cell Signaling: *A Preview*
- **CONCEPT 11.2 Signal reception: A signaling molecule binds to a receptor, causing it to change shape**
- ◦ Receptors in the Plasma Membrane
- Intracellular Receptors
- **CONCEPT 11.3 Signal transduction: Cascades of molecular interactions transmit signals from receptors to relay molecules in the cell**
- ◦ Signal Transduction Pathways
- Protein Phosphorylation and Dephosphorylation
- Small Molecules and Ions as Second Messengers
- **CONCEPT 11.4 Cellular response: Cell signaling leads to regulation of transcription or cytoplasmic activities**

- ◦ Nuclear and Cytoplasmic Responses
- Regulation of the Response
- **CONCEPT 11.5 Apoptosis requires integration of multiple cell-signaling pathways**
- ◦ Apoptosis in the Soil Worm *Caenorhabditis elegans*
- Apoptotic Pathways and the Signals That Trigger Them


## 12 The Cell Cycle


- **CONCEPT 12.1 Most cell division results in genetically identical daughter cells**
- ◦ Key Roles of Cell Division
- Cellular Organization of the Genetic Material
- Distribution of Chromosomes During Eukaryotic Cell Division
- **CONCEPT 12.2 The mitotic phase alternates with interphase in the cell cycle**
- ◦ Phases of the Cell Cycle
- The Mitotic Spindle: *A Closer Look*
- Cytokinesis: *A Closer Look*
- Binary Fission in Bacteria
- The Evolution of Mitosis
- **CONCEPT 12.3 The eukaryotic cell cycle is regulated by a molecular control system**
- ◦ The Cell Cycle Control System
- Loss of Cell Cycle Controls in Cancer Cells


## Unit 3 Genetics

**Interview: Francisco Mojica**

## 13 Meiosis and Sexual Life Cycles


- **CONCEPT 13.1 Offspring acquire genes from parents by inheriting chromosomes**

- - Inheritance of Genes
- Comparison of Asexual and Sexual Reproduction
- **CONCEPT 13.2 Fertilization and meiosis alternate in sexual life cycles**
- - Sets of Chromosomes in Human Cells
- Behavior of Chromosome Sets in the Human Life Cycle
- The Variety of Sexual Life Cycles
- **CONCEPT 13.3 Meiosis reduces the number of chromosome sets from diploid to haploid**
- - The Stages of Meiosis
- Crossing Over and Synapsis During Prophase I
- A Comparison of Mitosis and Meiosis
- **CONCEPT 13.4 Genetic variation produced in sexual life cycles contributes to evolution**
- - Origins of Genetic Variation Among Offspring
- The Evolutionary Significance of Genetic Variation Within Populations

# 14 Mendel and the Gene Idea

- **CONCEPT 14.1 Mendel used the scientific approach to identify two laws of inheritance**
- - Mendel's Experimental, Quantitative Approach
- The Law of Segregation
- The Law of Independent Assortment
- **CONCEPT 14.2 Probability laws govern Mendelian inheritance**
- - The Multiplication and Addition Rules Applied to Monohybrid Crosses
- Solving Complex Genetics Problems with the Rules of Probability
- **CONCEPT 14.3 Inheritance patterns are often more complex than predicted by simple Mendelian genetics**

- - Extending Mendelian Genetics for a Single Gene
- Extending Mendelian Genetics for Two or More Genes
- Nature and Nurture: The Environmental Impact on Phenotype
- A Mendelian View of Heredity and Variation
- **CONCEPT 14.4 Many human traits follow Mendelian patterns of inheritance**
- - Pedigree Analysis
- Recessively Inherited Disorders
- Dominantly Inherited Disorders
- Multifactorial Disorders
- Genetic Testing and Counseling

## 15 The Chromosomal Basis of Inheritance

- **CONCEPT 15.1 Mendelian inheritance has its physical basis in the behavior of chromosomes**
- - Morgan's Choice of Experimental Organism
- Correlating Behavior of a Gene's Alleles with Behavior of a Chromosome Pair: *Scientific Inquiry*
- **CONCEPT 15.2 Sex-linked genes exhibit unique patterns of inheritance**
- - The Chromosomal Basis of Sex
- Inheritance of X-Linked Genes
- X Inactivation in Female Mammals
- **CONCEPT 15.3 Linked genes tend to be inherited together because they are located near each other on the same chromosome**
- - How Linkage Affects Inheritance
- Genetic Recombination and Linkage
- Mapping the Distance Between Genes Using Recombination Data: *Scientific Inquiry*
- **CONCEPT 15.4 Alterations of chromosome number or structure cause some genetic disorders**
- - Abnormal Chromosome Number

- Alterations of Chromosome Structure
- Human Conditions Due to Chromosomal Alterations
- **CONCEPT 15.5 Some inheritance patterns are exceptions to standard Mendelian inheritance**
- ◦ Genomic Imprinting
- Inheritance of Organelle Genes

## 16 The Molecular Basis of Inheritance

- **CONCEPT 16.1 DNA is the genetic material**
- ◦ The Search for the Genetic Material: *Scientific Inquiry*
- Building a Structural Model of DNA
- **CONCEPT 16.2 Many proteins work together in DNA replication and repair**
- ◦ The Basic Principle: Base Pairing to a Template Strand
- DNA Replication: *A Closer Look*
- Proofreading and Repairing DNA
- Evolutionary Significance of Altered DNA Nucleotides
- Replicating the Ends of DNA Molecules
- **CONCEPT 16.3 A chromosome consists of a DNA molecule packed together with proteins**

## 17 Gene Expression: From Gene to Protein

- **CONCEPT 17.1 Genes specify proteins via transcription and translation**
- ◦ Evidence from Studying Metabolic Defects
- Basic Principles of Transcription and Translation
- The Genetic Code
- **CONCEPT 17.2 Transcription is the DNA-directed synthesis of RNA:**_**A Closer Look**_
- ◦ Molecular Components of Transcription

- Synthesis of an RNA Transcript
- **CONCEPT 17.3 Eukaryotic cells modify RNA after transcription**
- ◦ Alteration of mRNA Ends
- Split Genes and RNA Splicing
- **CONCEPT 17.4 Translation is the RNA-directed synthesis of a polypeptide:**_**A Closer Look**_
- ◦ Molecular Components of Translation
- Building a Polypeptide
- Completing and Targeting the Functional Protein
- Making Multiple Polypeptides in Bacteria and Eukaryotes
- **CONCEPT 17.5 Mutations of one or a few nucleotides can affect protein structure and function**
- ◦ Types of Small-Scale Mutations
- New Mutations and Mutagens
- Using CRISPR to Edit Genes and Correct Disease-Causing Mutations
- What Is a Gene? *Revisiting the Question*

## 18 Regulation of Gene Expression

- **CONCEPT 18.1 Bacteria often respond to environmental change by regulating transcription**
- ◦ Operons: The Basic Concept
- Repressible and Inducible Operons: Two Types of Negative Gene Regulation
- Positive Gene Regulation
- **CONCEPT 18.2 Eukaryotic gene expression is regulated at many stages**
- ◦ Differential Gene Expression
- Regulation of Chromatin Structure
- Regulation of Transcription Initiation
- Mechanisms of Post-transcriptional Regulation
- **CONCEPT 18.3 Noncoding RNAs play multiple roles in controlling gene expression**

- ◦ Effects on mRNAs by MicroRNAs and Small Interfering RNAs
- Chromatin Remodeling and Effects on Transcription by ncRNAs
- **CONCEPT 18.4 A program of differential gene expression leads to the different cell types in a multicellular organism**
- ◦ A Genetic Program for Embryonic Development
- Cytoplasmic Determinants and Inductive Signals
- Sequential Regulation of Gene Expression During Cellular Differentiation
- Pattern Formation: Setting Up the Body Plan
- **CONCEPT 18.5 Cancer results from genetic changes that affect cell cycle control**
- ◦ Types of Genes Associated with Cancer
- Interference with Normal Cell-Signaling Pathways
- The Multistep Model of Cancer Development
- Inherited Predisposition and Environmental Factors Contributing to Cancer
- The Role of Viruses in Cancer

# 19 Viruses

- **CONCEPT 19.1 A virus consists of a nucleic acid surrounded by a protein coat**
- ◦ The Discovery of Viruses: *Scientific Inquiry*
- Structure of Viruses
- **CONCEPT 19.2 Viruses replicate only in host cells**
- ◦ General Features of Viral Replicative Cycles
- Replicative Cycles of Phages
- Replicative Cycles of Animal Viruses
- Evolution of Viruses
- **CONCEPT 19.3 Viruses and prions are formidable pathogens in animals and plants**
- ◦ Viral Diseases in Animals
- Emerging Viral Diseases

- Viral Diseases in Plants
- Prions: Proteins as Infectious Agents

## 20 DNA Tools and Biotechnology

- **CONCEPT 20.1 DNA sequencing and DNA cloning are valuable tools for genetic engineering and biological inquiry**
- ◦ DNA Sequencing
- Making Multiple Copies of a Gene or Other DNA Segment
- Using Restriction Enzymes to Make a Recombinant DNA Plasmid
- Amplifying DNA: The Polymerase Chain Reaction (PCR) and Its Use in DNA Cloning
- Expressing Cloned Eukaryotic Genes
- **CONCEPT 20.2 Biologists use DNA technology to study gene expression and function**
- ◦ Analyzing Gene Expression
- Determining Gene Function
- **CONCEPT 20.3 Cloned organisms and stem cells are useful for basic research and other applications**
- ◦ Cloning Plants: Single-Cell Cultures
- Cloning Animals: Nuclear Transplantation
- Stem Cells of Animals
- **CONCEPT 20.4 The practical applications of DNA-based biotechnology affect our lives in many ways**
- ◦ Medical Applications
- Forensic Evidence and Genetic Profiles
- Environmental Cleanup
- Agricultural Applications
- Safety and Ethical Questions Raised by DNA Technology

## 21 Genomes and Their Evolution

- **CONCEPT 21.1 The Human Genome Project fostered development of faster, less expensive sequencing techniques**
- **CONCEPT 21.2 Scientists use bioinformatics to analyze genomes and their functions**
- ◦ Centralized Resources for Analyzing Genome Sequences
- Identifying Protein-Coding Genes and Understanding Their Functions
- Understanding Genes and Gene Expression at the Systems Level
- **CONCEPT 21.3 Genomes vary in size, number of genes, and gene density**
- ◦ Genome Size
- Number of Genes
- Gene Density and Noncoding DNA
- **CONCEPT 21.4 Multicellular eukaryotes have a lot of noncoding DNA and many multigene families**
- ◦ Transposable Elements and Related Sequences
- Other Repetitive DNA, Including Simple Sequence DNA
- Genes and Multigene Families
- **CONCEPT 21.5 Duplication, rearrangement, and mutation of DNA contribute to genome evolution**
- ◦ Duplication of Entire Chromosome Sets
- Alterations of Chromosome Structure
- Duplication and Divergence of Gene-Sized Regions of DNA
- Rearrangements of Parts of Genes: Exon Duplication and Exon Shuffling
- How Transposable Elements Contribute to Genome Evolution
- **CONCEPT 21.6 Comparing genome sequences provides clues to evolution and development**
- ◦ Comparing Genomes
- Widespread Conservation of Developmental Genes Among Animals

# Unit 4 Mechanisms of Evolution

**Interview: Cassandra Extavour**

## 22 Descent with Modification: A Darwinian View of Life

- **CONCEPT 22.1 The Darwinian revolution challenged traditional views of a young Earth inhabited by unchanging species**
- ◦ Endless Forms Most Beautiful
- *Scala Naturae* and Classification of Species
- Ideas About Change over Time
- Lamarck's Hypothesis of Evolution
- **CONCEPT 22.2 Descent with modification by natural selection explains the adaptations of organisms and the unity and diversity of life**
- ◦ Darwin's Research
- Ideas from*The Origin of Species*
- Key Features of Natural Selection
- **CONCEPT 22.3 Evolution is supported by an overwhelming amount of scientific evidence**
- ◦ Direct Observations of Evolutionary Change
- Homology
- The Fossil Record
- Biogeography
- What Is Theoretical About Darwin's View of Life?

## 23 The Evolution of Populations

- **CONCEPT 23.1 Genetic variation makes evolution possible**
- ◦ Genetic Variation
- Sources of Genetic Variation
- **CONCEPT 23.2 The Hardy-Weinberg equation can be used to test whether a population is evolving**

- ◦ Gene Pools and Allele Frequencies
- The Hardy-Weinberg Equation
- **CONCEPT 23.3 Natural selection, genetic drift, and gene flow can alter allele frequencies in a population**
- ◦ Natural Selection
- Genetic Drift
- Gene Flow
- **CONCEPT 23.4 Natural selection is the only mechanism that consistently causes adaptive evolution**
- ◦ Natural Selection: *A Closer Look*
- The Key Role of Natural Selection in Adaptive Evolution
- Sexual Selection
- Balancing Selection
- Why Natural Selection Cannot Fashion Perfect Organisms

## 24 The Origin of Species

- **CONCEPT 24.1 The biological species concept emphasizes reproductive isolation**
- ◦ The Biological Species Concept
- ◦ Other Definitions of Species
- **CONCEPT 24.2 Speciation can take place with or without geographic separation**
- ◦ Allopatric (“Other Country”) Speciation
- ◦ Sympatric (“Same Country”) Speciation
- ◦ Allopatric and Sympatric Speciation:*A Review*
- **CONCEPT 24.3 Hybrid zones reveal factors that cause reproductive isolation**
- ◦ Patterns Within Hybrid Zones

- Hybrid Zones and Environmental Change
- Hybrid Zones over Time
- **CONCEPT 24.4 Speciation can occur rapidly or slowly and can result from changes in few or many genes**
- ◦ The Time Course of Speciation
- Studying the Genetics of Speciation
- From Speciation to Macroevolution

# 25 The History of Life on Earth

- **CONCEPT 25.1 Conditions on early Earth made the origin of life possible**
- ◦ Synthesis of Organic Compounds on Early Earth
- Abiotic Synthesis of Macromolecules
- Protocells
- Self-Replicating RNA
- **CONCEPT 25.2 The fossil record documents the history of life**
- ◦ The Fossil Record
- How Rocks and Fossils Are Dated
- The Origin of New Groups of Organisms
- **CONCEPT 25.3 Key events in life's history include the origins of unicellular and multicellular organisms and the colonization of land**
- ◦ The First Single-Celled Organisms
- The Origin of Multicellularity
- The Colonization of Land
- **CONCEPT 25.4 The rise and fall of groups of organisms reflect differences in speciation and extinction rates**
- ◦ Plate Tectonics
- Mass Extinctions
- Adaptive Radiations

- **CONCEPT 25.5 Major changes in body form can result from changes in the sequences and regulation of developmental genes**
- ◦ Effects of Developmental Genes
- The Evolution of Development
- **CONCEPT 25.6 Evolution is not goal oriented**
- ◦ Evolutionary Novelties
- Evolutionary Trends

# Unit 5 The Evolutionary History

of Biological Diversity

**Interview: Penny Chisholm**

# 26 Phylogeny and the Tree of Life

- **CONCEPT 26.1 Phylogenies show evolutionary relationships**
- ◦ Binomial Nomenclature
- Hierarchical Classification
- Linking Classification and Phylogeny
- What We Can and Cannot Learn from Phylogenetic Trees
- Applying Phylogenies
- **CONCEPT 26.2 Phylogenies are inferred from morphological and molecular data**
- ◦ Morphological and Molecular Homologies
- Sorting Homology from Analogy
- Evaluating Molecular Homologies
- **CONCEPT 26.3 Shared characters are used to construct phylogenetic trees**
- ◦ Cladistics
- Phylogenetic Trees with Proportional Branch Lengths
- Maximum Parsimony and Maximum Likelihood

- Phylogenetic Trees as Hypotheses
- **CONCEPT 26.4 An organism's evolutionary history is documented in its genome**
- ◦ Gene Duplications and Gene Families
- ◦ Genome Evolution
- **CONCEPT 26.5 Molecular clocks help track evolutionary time**
- ◦ Molecular Clocks
- ◦ Applying a Molecular Clock: Dating the Origin of HIV
- **CONCEPT 26.6 Our understanding of the tree of life continues to change based on new data**
- ◦ From Two Kingdoms to Three Domains
- ◦ The Important Role of Horizontal Gene Transfer

## 27 Bacteria and Archaea

- **CONCEPT 27.1 Structural and functional adaptations contribute to prokaryotic success**
- ◦ Cell-Surface Structures
- Motility
- Internal Organization and DNA
- Reproduction
- **CONCEPT 27.2 Rapid reproduction, mutation, and genetic recombination promote genetic diversity in prokaryotes**
- ◦ Rapid Reproduction and Mutation
- Genetic Recombination
- **CONCEPT 27.3 Diverse nutritional and metabolic adaptations have evolved in prokaryotes**
- ◦ The Role of Oxygen in Metabolism

- Nitrogen Metabolism
- Metabolic Cooperation
- **CONCEPT 27.4 Prokaryotes have radiated into a diverse set of lineages**
- ◦ An Overview of Prokaryotic Diversity
- Bacteria
- Archaea
- **CONCEPT 27.5 Prokaryotes play crucial roles in the biosphere**
- ◦ Chemical Recycling
- Ecological Interactions
- **CONCEPT 27.6 Prokaryotes have both beneficial and harmful impacts on humans**
- ◦ Mutualistic Bacteria
- Pathogenic Bacteria
- Antibiotic Resistance
- Prokaryotes in Research and Technology

## 28 Protists

- **CONCEPT 28.1 Most eukaryotes are single-celled organisms**
- ◦ Structural and Functional Diversity in Protists
- Endosymbiosis in Eukaryotic Evolution
- Four Supergroups of Eukaryotes
- **CONCEPT 28.2 Excavates include protists with modified mitochondria and protists with unique flagella**
- ◦ Diplomonads and Parabasalids
- Euglenozoans
- **CONCEPT 28.3 SAR is a highly diverse group of protists defined by DNA similarities**
- ◦ Stramenopiles
- Alveolates

- Rhizarians
- **CONCEPT 28.4 Red algae and green algae are the closest relatives of plants**
- ◦ Red Algae
- Green Algae
- **CONCEPT 28.5 Unikonts include protists that are closely related to fungi and animals**
- ◦ Amoebozoans
- Opisthokonts
- **CONCEPT 28.6 Protists play key roles in ecological communities**
- ◦ Symbiotic Protists
- Photosynthetic Protists

# 29 Plant Diversity I: How Plants Colonized Land

- **CONCEPT 29.1 Plants evolved from green algae**
- ◦ Evidence of Algal Ancestry
- Adaptations Enabling the Move to Land
- Derived Traits of Plants
- The Origin and Diversification of Plants
- **CONCEPT 29.2 Mosses and other nonvascular plants have life cycles dominated by gametophytes**
- ◦ Bryophyte Gametophytes
- Bryophyte Sporophytes
- The Ecological and Economic Importance of Mosses
- **CONCEPT 29.3 Ferns and other seedless vascular plants were the first plants to grow tall**
- ◦ Origins and Traits of Vascular Plants
- Classification of Seedless Vascular Plants

- The Significance of Seedless Vascular Plants

# 30 Plant Diversity II: The Evolution of Seed Plants

- **CONCEPT 30.1 Seeds and pollen grains are key adaptations for life on land**
- ◦ Advantages of Reduced Gametophytes
- Heterospory: The Rule Among Seed Plants
- Ovules and Production of Eggs
- Pollen and Production of Sperm
- The Evolutionary Advantage of Seeds
- **CONCEPT 30.2 Gymnosperms bear “naked” seeds, typically on cones**
- ◦ The Life Cycle of a Pine
- Early Seed Plants and the Rise of Gymnosperms
- Gymnosperm Diversity
- **CONCEPT 30.3 The reproductive adaptations of angiosperms include flowers and fruits**
- ◦ Characteristics of Angiosperms
- Angiosperm Evolution
- Angiosperm Diversity
- **CONCEPT 30.4 Human welfare depends on seed plants**
- ◦ Products from Seed Plants
- Threats to Plant Diversity

# 31 Fungi

- **CONCEPT 31.1 Fungi are heterotrophs that feed by absorption**
- ◦ Nutrition and Ecology
- Body Structure
- Specialized Hyphae in Mycorrhizal Fungi

- **CONCEPT 31.2 Fungi produce spores through sexual or asexual life cycles**
- ◦ Sexual Reproduction
- Asexual Reproduction
- **CONCEPT 31.3 The ancestor of fungi was an aquatic, single-celled, flagellated protist**
- ◦ The Origin of Fungi
- The Move to Land
- **CONCEPT 31.4 Fungi have radiated into a diverse set of lineages**
- ◦ Cryptomycetes and Microsporidians
- Zoopagomycetes
- Mucoromycetes
- Ascomycetes
- Basidiomycetes
- **CONCEPT 31.5 Fungi play key roles in nutrient cycling, ecological interactions, and human welfare**
- ◦ Fungi as Decomposers
- Fungi as Mutualists
- Practical Uses of Fungi

## 32 An Overview of Animal Diversity

- **CONCEPT 32.1 Animals are multicellular, heterotrophic eukaryotes with tissues that develop from embryonic layers**
- ◦ Nutritional Mode
- Cell Structure and Specialization
- Reproduction and Development
- **CONCEPT 32.2 The history of animals spans more than half a billion years**
- ◦ Steps in the Origin of Multicellular Animals
- Neoproterozoic Era (1 Billion–541 Million Years Ago)

- Paleozoic Era (541–252 Million Years Ago)
- Mesozoic Era (252–66 Million Years Ago)
- Cenozoic Era (66 Million Years Ago to the Present)
- **CONCEPT 32.3 Animals can be characterized by body plans**
- ◦ Symmetry

- Tissues
- Body Cavities
- Protostome and Deuterostome Development
- **CONCEPT 32.4 Views of animal phylogeny continue to be shaped by new molecular and morphological data**
- ◦ The Diversification of Animals

- Future Directions in Animal Systematics

## 33 An Introduction to Invertebrates

- **CONCEPT 33.1 Sponges are basal animals that lack tissues**
- **CONCEPT 33.2 Cnidarians are an ancient phylum of eumetazoans**
- ◦ Medusozoans

- Anthozoans
- **CONCEPT 33.3 Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms**
- ◦ Flatworms

- Rotifers and Acanthocephalans
- Lophophorates: Ectoprocts and Brachiopods
- Molluscs
- Annelids
- **CONCEPT 33.4 Ecdysozoans are the most species-rich animal group**
- ◦ Nematodes

- Arthropods

- **CONCEPT 33.5 Echinoderms and chordates are deuterostomes**
- - Echinoderms
- Chordates

# 34 The Origin and Evolution of Vertebrates

- **CONCEPT 34.1 Chordates have a notochord and a dorsal, hollow nerve cord**
- - Derived Characters of Chordates
- Lancelets
- Tunicates
- Early Chordate Evolution
- **CONCEPT 34.2 Vertebrates are chordates that have a backbone**
- - Derived Characters of Vertebrates
- Hagfishes and Lampreys
- Early Vertebrate Evolution
- **CONCEPT 34.3 Gnathostomes are vertebrates that have jaws**
- - Derived Characters of Gnathostomes
- Fossil Gnathostomes
- Chondrichthyans (Sharks, Rays, and Their Relatives)
- Ray-Finned Fishes and Lobe-Fins
- **CONCEPT 34.4 Tetrapods are gnathostomes that have limbs**
- - Derived Characters of Tetrapods
- The Origin of Tetrapods
- Amphibians
- **CONCEPT 34.5 Amniotes are tetrapods that have a terrestrially adapted egg**
- - Derived Characters of Amniotes
- Early Amniotes
- Reptiles

- **CONCEPT 34.6 Mammals are amniotes that have hair and produce milk**
- - Derived Characters of Mammals
- Early Evolution of Mammals
- Monotremes
- Marsupials
- Eutherians (Placental Mammals)
- **CONCEPT 34.7 Humans are mammals that have a large brain and bipedal locomotion**
- - Derived Characters of Humans
- The Earliest Hominins
- Australopiths
- Bipedalism
- Tool Use
- Early *Homo*
- - Neanderthals
- *Homo sapiens*

# Unit 6 Plant Form and Function

**Interview: Dennis Gonsalves**

# 35 Vascular Plant Structure, Growth,

and Development

- **CONCEPT 35.1 Plants have a hierarchical organization consisting of organs, tissues, and cells**
- - Vascular Plant Organs: Roots, Stems, and Leaves
- Dermal, Vascular, and Ground Tissues
- Common Types of Plant Cells

- **CONCEPT 35.2 Different meristems generate new cells for primary and secondary growth**
- **CONCEPT 35.3 Primary growth lengthens roots and shoots**
- ◦ Primary Growth of Roots
- Primary Growth of Shoots
- **CONCEPT 35.4 Secondary growth increases the diameter of stems and roots in woody plants**
- ◦ The Vascular Cambium and Secondary Vascular Tissue
- The Cork Cambium and the Production of Periderm
- Evolution of Secondary Growth
- **CONCEPT 35.5 Growth, morphogenesis, and cell differentiation produce the plant body**
- ◦ Model Organisms: Revolutionizing the Study of Plants
- Growth: Cell Division and Cell Expansion
- Morphogenesis and Pattern Formation
- Gene Expression and the Control of Cell Differentiation
- Shifts in Development: Phase Changes
- Genetic Control of Flowering

## 36 Resource Acquisition and Transport in Vascular Plants

- **CONCEPT 36.1 Adaptations for acquiring resources were key steps in the evolution of vascular plants**
- ◦ Shoot Architecture and Light Capture
- Root Architecture and Acquisition of Water and Minerals
- **CONCEPT 36.2 Different mechanisms transport substances over short or long distances**
- ◦ The Apoplast and Symplast: Transport Continuums
- Short-Distance Transport of Solutes Across Plasma Membranes

- Short-Distance Transport of Water Across Plasma Membranes
- Long-Distance Transport:The Role of Bulk Flow
- **CONCEPT 36.3 Transpiration drives the transport of water and minerals from roots to shoots via the xylem**
- ◦ Absorption of Water and Minerals by Root Cells
- Transport of Water and Minerals into the Xylem
- Bulk Flow Transport via the Xylem
- Xylem Sap Ascent by Bulk Flow: *A Review*
- **CONCEPT 36.4 The rate of transpiration is regulated by stomata**
- ◦ Stomata: Major Pathways for Water Loss
- Mechanisms of Stomatal Opening and Closing
- Stimuli for Stomatal Opening and Closing
- Effects of Transpiration on Wilting and Leaf Temperature
- Adaptations That Reduce Evaporative Water Loss
- **CONCEPT 36.5 Sugars are transported from sources to sinks via the phloem**
- ◦ Movement from Sugar Sources to Sugar Sinks
- Bulk Flow by Positive Pressure: The Mechanism of Translocation in Angiosperms
- **CONCEPT 36.6 The symplast is highly dynamic**
- ◦ Changes in Plasmodesmatal Number and Pore Size
- Phloem: An Information Superhighway
- Electrical Signaling in the Phloem

# 37 Soil and Plant Nutrition

- **CONCEPT 37.1 Soil contains a living, complex ecosystem**
- ◦ Soil Texture
- Topsoil Composition
- Soil Conservation and Sustainable Agriculture
- **CONCEPT 37.2 Plant roots absorb many types of essential elements from the soil**

- - Essential Elements
- Symptoms of Mineral Deficiency
- Global Climate Change and Food Quality
- **CONCEPT 37.3 Plant nutrition often involves relationships with other organisms**
- - Bacteria and Plant Nutrition
- Fungi and Plant Nutrition
- Epiphytes, Parasitic Plants, and Carnivorous Plants

# 38 Angiosperm Reproduction and Biotechnology

- **CONCEPT 38.1 Flowers, double fertilization, and fruits are key features of the angiosperm life cycle**
- - Flower Structure and Function
- Methods of Pollination
- The Angiosperm Life Cycle: An Overview
- Development of Female Gametophytes (Embryo Sacs)
- Development of Male Gametophytes in Pollen Grains
- Seed Development and Structure
- Sporophyte Development from Seed to Mature Plant
- Fruit Structure and Function
- **CONCEPT 38.2 Flowering plants reproduce sexually, asexually, or both**
- - Mechanisms of Asexual Reproduction
- Advantages and Disadvantages of Asexual and Sexual Reproduction
- Mechanisms That Prevent Self-Fertilization
- Totipotency, Vegetative Reproduction, and Tissue Culture
- **CONCEPT 38.3 People modify crops by breeding and genetic engineering**
- - Plant Breeding
- Plant Biotechnology and Genetic Engineering

- The Debate over Plant Biotechnology

# 39 Plant Responses to Internal

and External Signals

- **CONCEPT 39.1 Signal transduction pathways link signal reception to response**
- ◦ Reception

- Transduction
- Response
- **CONCEPT CHECK 39.2 Plants use chemicals to communicate**
- ◦ General Characteristics of Plant Hormones

- A Survey of Plant Hormones
- **CONCEPT 39.3 Responses to light are critical for plant success**
- ◦ Blue-Light Photoreceptors

- Phytochrome Photoreceptors
- Biological Clocks and Circadian Rhythms
- The Effect of Light on the Biological Clock
- Photoperiodism and Responses to Seasons
- **CONCEPT 39.4 Plants respond to a wide variety of stimuli other than light**
- ◦ Gravity

- Mechanical Stimuli
- Environmental Stresses
- **CONCEPT 39.5 Plants respond to attacks by pathogens and herbivores**
- ◦ Defenses Against Pathogens

- Defenses Against Herbivores

# Unit 7 Animal Form and Function

**Interview: Steffanie Strathdee**

# 40 Basic Principles of Animal Form

and Function

- **CONCEPT 40.1 Animal form and function are correlated at all levels of organization**
- ◦ Evolution of Animal Size and Shape
- Exchange with the Environment
- Hierarchical Organization of Body Plans
- Coordination and Control
- **CONCEPT 40.2 Feedback control maintains the internal environment in many animals**
- ◦ Regulating and Conforming
- Homeostasis
- **CONCEPT 40.3 Homeostatic processes for thermoregulation involve form, function, and behavior**
- ◦ Endothermy and Ectothermy
- Variation in Body Temperature
- Balancing Heat Loss and Gain
- Acclimatization in Thermoregulation
- Physiological Thermostats and Fever
- **CONCEPT 40.4 Energy requirements are related to animal size, activity, and environment**
- ◦ Energy Allocation and Use
- Quantifying Energy Use
- Minimum Metabolic Rate and Thermoregulation
- Influences on Metabolic Rate
- Torpor and Energy Conservation

# 41 Animal Nutrition

- **CONCEPT 41.1 An animal's diet must supply chemical energy, organic building blocks, and essential nutrients**
- ◦ Essential Nutrients
- Variation in Diet
- Dietary Deficiencies
- Assessing Nutritional Needs
- **CONCEPT 41.2 Food processing involves ingestion, digestion, absorption, and elimination**
- ◦ Digestive Compartments
- **CONCEPT 41.3 Organs specialized for sequential stages of food processing form the mammalian digestive system**
- ◦ The Oral Cavity, Pharynx, and Esophagus
- Digestion in the Stomach
- Digestion in the Small Intestine
- Absorption in the Small Intestine
- Processing in the Large Intestine
- **CONCEPT 41.4 Evolutionary adaptations of vertebrate digestive systems correlate with diet**
- ◦ Dental Adaptations
- Stomach and Intestinal Adaptations
- Mutualistic Adaptations
- **CONCEPT 41.5 Feedback circuits regulate digestion, energy storage, and appetite**
- ◦ Regulation of Digestion
- Regulation of Energy Storage
- Regulation of Appetite and Consumption

# 42 Circulation and Gas Exchange

- **CONCEPT 42.1 Circulatory systems link exchange surfaces with cells throughout the body**
- ◦ Gastrovascular Cavities
- Open and Closed Circulatory Systems
- Organization of Vertebrate Circulatory Systems
- **CONCEPT 42.2 Coordinated cycles of heart contraction drive double circulation in mammals**
- ◦ Mammalian Circulation
- The Mammalian Heart: *A Closer Look*
- Maintaining the Heart's Rhythmic Beat
- **CONCEPT 42.3 Patterns of blood pressure and flow reflect the structure and arrangement of blood vessels**
- ◦ Blood Vessel Structure and Function
- Blood Flow Velocity
- Blood Pressure
- Capillary Function
- Fluid Return by the Lymphatic System
- **CONCEPT 42.4 Blood components function in exchange, transport, and defense**
- ◦ Blood Composition and Function
- Cardiovascular Disease
- **CONCEPT 42.5 Gas exchange occurs across specialized respiratory surfaces**
- ◦ Partial Pressure Gradients in Gas Exchange
- Respiratory Media
- Respiratory Surfaces
- Gills in Aquatic Animals
- Tracheal Systems in Insects
- Lungs

- **CONCEPT 42.6 Breathing ventilates the lungs**
- ◦ How an Amphibian Breathes
- How a Bird Breathes
- How a Mammal Breathes
- Control of Breathing in Humans
- **CONCEPT 42.7 Adaptations for gas exchange include pigments that bind and transport gases**
- ◦ Coordination of Circulation and Gas Exchange
- Respiratory Pigments
- Respiratory Adaptations of Diving Mammals

# 43 The Immune System

- **CONCEPT 43.1 In innate immunity, recognition and response rely on traits common to groups of pathogens**
- ◦ Innate Immunity of Invertebrates
- Innate Immunity of Vertebrates
- Evasion of Innate Immunity by Pathogens
- **CONCEPT 43.2 In adaptive immunity, receptors provide pathogen-specific recognition**
- ◦ Antigens as the Trigger for Adaptive Immunity
- Antigen Recognition by B Cells and Antibodies
- Antigen Recognition by T Cells
- B Cell and T Cell Development
- **CONCEPT 43.3 Adaptive immunity defends against infection of body fluids and body cells**
- ◦ Helper T Cells: Activating Adaptive Immunity
- B Cells and Antibodies: A Response to Extracellular Pathogens
- Cytotoxic T Cells: A Response to Infected Host Cells

- Summary of the Humoral and Cell-Mediated Immune Responses
- Immunization
- Active and Passive Immunity
- Antibodies as Tools
- Immune Rejection
- **CONCEPT 43.4 Disruptions in immune system function can elicit or exacerbate disease**
- ◦ Exaggerated, Self-Directed, and Diminished Immune Responses
- Evolutionary Adaptations of Pathogens That Underlie Immune System Avoidance
- Cancer and Immunity

# 44 Osmoregulation and Excretion

- **CONCEPT 44.1 Osmoregulation balances the uptake and loss of water and solutes**
- ◦ Osmosis and Osmolarity
- Osmoregulatory Challenges and Mechanisms
- Energetics of Osmoregulation
- Transport Epithelia in Osmoregulation
- **CONCEPT 44.2 An animal's nitrogenous wastes reflect its phylogeny and habitat**
- ◦ Forms of Nitrogenous Waste
- The Influence of Evolution and Environment on Nitrogenous Wastes
- **CONCEPT 44.3 Diverse excretory systems are variations on a tubular theme**
- ◦ Survey of Excretory Systems
- **CONCEPT 44.4 The nephron is organized for stepwise processing of blood filtrate**
- ◦ From Blood Filtrate to Urine: *A Closer Look*
- Solute Gradients and Water Conservation
- Adaptations of the Vertebrate Kidney to Diverse Environments
- **CONCEPT 44.5 Hormonal circuits link kidney function, water balance, and blood pressure**

- ◦ Homeostatic Regulation of the Kidney

# 45 Hormones and the Endocrine System

- **CONCEPT 45.1 Hormones and other signaling molecules bind to target receptors, triggering specific response pathways**
- ◦ Intercellular Information Flow

- Chemical Classes of Hormones
- Cellular Hormone Response Pathways
- Endocrine Tissues and Organs
- **CONCEPT 45.2 Feedback regulation and coordination with the nervous system are common in hormone pathways**
- ◦ Simple Endocrine Pathways

- Simple Neuroendocrine Pathways
- Feedback Regulation
- Coordination of the Endocrine and Nervous Systems
- Thyroid Regulation: A Hormone Cascade Pathway
- Hormonal Regulation of Growth
- **CONCEPT 45.3 Endocrine glands respond to diverse stimuli in regulating homeostasis, development, and behavior**
- ◦ Parathyroid Hormone and Vitamin D: Control of Blood Calcium

- Adrenal Hormones: Response to Stress
- Sex Hormones
- Hormones and Biological Rhythms
- Evolution of Hormone Function

# 46 Animal Reproduction

- **CONCEPT 46.1 Both asexual and sexual reproduction occur in the animal kingdom**

- ◦ Mechanisms of Asexual Reproduction
- Variation in Patterns of Sexual Reproduction
- Reproductive Cycles
- Sexual Reproduction: An Evolutionary Enigma
- **CONCEPT 46.2 Fertilization depends on mechanisms that bring together sperm and eggs of the same species**
- ◦ Ensuring the Survival of Offspring
- Gamete Production and Delivery
- **CONCEPT 46.3 Reproductive organs produce and transport gametes**
- ◦ Human Male Reproductive Anatomy
- Human Female Reproductive Anatomy
- Gametogenesis
- **CONCEPT 46.4 The interplay of tropic and sex hormones regulates reproduction in mammals**
- ◦ Biological Sex and Human Sexuality
- Hormonal Control of the Male Reproductive System
- Hormonal Control of Female Reproductive Cycles
- Human Sexual Response
- **CONCEPT 46.5 In placental mammals, an embryo develops fully within the mother's uterus**
- ◦ Conception, Embryonic Development, and Birth
- Maternal Immune Tolerance of the Embryo and Fetus
- Contraception and Abortion
- Modern Reproductive Technologies

# 47 Animal Development

- **CONCEPT 47.1 Fertilization and cleavage initiate embryonic development**
- ◦ Fertilization

- Cleavage
- **CONCEPT 47.2 Morphogenesis in animals involves specific changes in cell shape, position, and survival**
- ◦ Gastrulation
- Developmental Adaptations of Amniotes
- Organogenesis
- The Cytoskeleton in Morphogenesis
- **CONCEPT 47.3 Cytoplasmic determinants and inductive signals regulate cell fate**
- ◦ Fate Mapping
- Axis Formation
- Restricting Developmental Potential
- Cell Fate Determination and Pattern Formation by Inductive Signals
- Cilia and Cell Fate

## 48 Neurons, Synapses, and

Signaling

- **CONCEPT 48.1 Neuron structure and organization reflect function in information transfer**
- ◦ Neuron Structure and Function
- Introduction to Information Processing
- **CONCEPT 48.2 Ion pumps and ion channels establish the resting potential of a neuron**
- ◦ Formation of the Resting Potential
- Modeling the Resting Potential
- **CONCEPT 48.3 Action potentials are the signals conducted by axons**
- ◦ Hyperpolarization and Depolarization
- Graded Potentials and Action Potentials
- Generation of Action Potentials: *A Closer Look*

- Conduction of Action Potentials
- **CONCEPT 48.4 Neurons communicate with other cells at synapses**
- ◦ Generation of Postsynaptic Potentials
- Summation of Postsynaptic Potentials
- Termination of Neurotransmitter Signaling
- Modulated Signaling at Synapses
- Neurotransmitters

# 49 Nervous Systems

- **CONCEPT 49.1 Nervous systems consist of circuits of neurons and supporting cells**
- ◦ Organization of the Vertebrate Nervous System
- The Peripheral Nervous System
- Glia
- **CONCEPT 49.2 The vertebrate brain is regionally specialized**
- ◦ Arousal and Sleep
- Biological Clock Regulation
- Emotions
- Functional Imaging of the Brain
- **CONCEPT 49.3 The cerebral cortex controls voluntary movement and cognitive functions**
- ◦ Information Processing
- Language and Speech
- Lateralization of Cortical Function
- Frontal Lobe Function
- Evolution of Cognition in Vertebrates
- **CONCEPT 49.4 Changes in synaptic connections underlie memory and learning**
- ◦ Neuronal Plasticity
- Memory and Learning

- Long-Term Potentiation
- **CONCEPT 49.5 Many nervous system disorders can now be explained in molecular terms**
- ◦ Schizophrenia
- Depression
- The Brain's Reward System and Drug Addiction
- Alzheimer's Disease
- Parkinson's Disease
- Future Directions in Brain Research

# 50 Sensory and Motor

Mechanisms

- **CONCEPT 50.1 Sensory receptors transduce stimulus energy and transmit signals to the central nervous system**
- ◦ Sensory Reception and Transduction
- Transmission
- Perception
- Amplification and Adaptation
- Types of Sensory Receptors
- **CONCEPT 50.2 In hearing and equilibrium, mechanoreceptors detect moving fluid or settling particles**
- ◦ Sensing of Gravity and Sound in Invertebrates
- Hearing and Equilibrium in Mammals
- Hearing and Equilibrium in Other Vertebrates
- **CONCEPT 50.3 The diverse visual receptors of animals depend on light-absorbing pigments**
- ◦ Evolution of Visual Perception
- The Vertebrate Visual System

- **CONCEPT 50.4 The senses of taste and smell rely on similar sets of sensory receptors**
- ◦ Taste in Mammals
- Smell in Humans
- **CONCEPT 50.5 The physical interaction of protein filaments is required for muscle function**
- ◦ Vertebrate Skeletal Muscle
- Other Types of Muscle
- **CONCEPT 50.6 Skeletal systems transform muscle contraction into locomotion**
- ◦ Types of Skeletal Systems
- Types of Locomotion

# 51 Animal Behavior

- **CONCEPT 51.1 Discrete sensory inputs can stimulate both simple and complex behaviors**
- ◦ Fixed Action Patterns
- Migration
- Behavioral Rhythms
- Animal Signals and Communication
- **CONCEPT 51.2 Learning establishes specific links between experience and behavior**
- ◦ Experience and Behavior
- Learning
- **CONCEPT 51.3 Selection for individual survival and reproductive success can explain diverse behaviors**
- ◦ Evolution of Foraging Behavior
- Mating Behavior and Mate Choice
- **CONCEPT 51.4 Genetic analyses and the concept of inclusive fitness provide a basis for studying the evolution of behavior**

- ◦ Genetic Basis of Behavior
- Genetic Variation and the Evolution of Behavior
- Altruism
- Inclusive Fitness
- Evolution and Human Culture

# Unit 8 Ecology

**Interview: Chelsea Rochman**

# 52 An Introduction to Ecology and the Biosphere

- **CONCEPT 52.1 Earth's climate varies by latitude and season and is changing rapidly**
- ◦ Global Climate Patterns
- Regional and Local Effects on Climate
- Effects of Vegetation on Climate
- Microclimate
- Global Climate Change
- **CONCEPT 52.2 The distribution of terrestrial biomes is controlled by climate and disturbance**
- ◦ Climate and Terrestrial Biomes
- General Features of Terrestrial Biomes
- Disturbance and Terrestrial Biomes
- **CONCEPT 52.3 Aquatic biomes are diverse and dynamic systems that cover most of Earth**
- ◦ Zonation in Aquatic Biomes
- **CONCEPT 52.4 Interactions between organisms and the environment limit the distribution of species**
- ◦ Dispersal and Distribution

- Biotic Factors
- Abiotic Factors
- **CONCEPT 52.5 Ecological change and evolution affect one another over long and short periods of time**

# 53 Population Ecology

- **CONCEPT 53.1 Biotic and abiotic factors affect population density, dispersion, and demographics**
- ◦ Density and Dispersion
- Demographics
- **CONCEPT 53.2 The exponential model describes population growth in an idealized, unlimited environment**
- ◦ Changes in Population Size
- Exponential Growth
- **CONCEPT 53.3 The logistic model describes how a population grows more slowly as it nears its carrying capacity**
- ◦ The Logistic Growth Model
- The Logistic Model and Real Populations
- **CONCEPT 53.4 Life history traits are products of natural selection**
- ◦ Diversity of Life Histories
- “Trade-offs” and Life Histories
- **CONCEPT 53.5 Density-dependent factors regulate population growth**
- ◦ Population Change and Population Density
- Mechanisms of Density-Dependent Population Regulation
- Population Dynamics
- **CONCEPT 53.6 The human population is no longer growing exponentially but is still increasing extremely rapidly**
- ◦ The Global Human Population

- Global Carrying Capacity

# 54 Community Ecology

- **CONCEPT 54.1 Interactions between species can help, harm, or have no effect on the individuals involved**
- ◦ Competition
- Exploitation
- Positive Interactions
- **CONCEPT 54.2 Diversity and trophic structure characterize biological communities**
- ◦ Species Diversity
- Diversity and Community Stability
- Trophic Structure
- Species with a Large Impact
- Bottom-Up and Top-Down Controls
- **CONCEPT 54.3 Disturbance influences species diversity and composition**
- ◦ Characterizing Disturbance
- Ecological Succession
- Human Disturbance
- **CONCEPT 54.4 Biogeographic factors affect community diversity**
- ◦ Latitudinal Gradients
- Area Effects
- Island Equilibrium Model
- **CONCEPT 54.5 Pathogens alter community structure locally and globally**
- ◦ Effects on Community Structure
- Community Ecology and Zoonotic Diseases

# 55 Ecosystems and Restoration Ecology

- **CONCEPT 55.1 Physical laws govern energy flow and chemical cycling in ecosystems**
- ◦ Energy Flow and Chemical Cycling
- Conservation of Energy
- Conservation of Mass
- Energy, Mass, and Trophic Levels
- **CONCEPT 55.2 Energy and other limiting factors control primary production in ecosystems**
- ◦ Ecosystem Energy Budgets
- Primary Production in Aquatic Ecosystems
- Primary Production in Terrestrial Ecosystems
- **CONCEPT 55.3 Energy transfer between trophic levels is typically only** 10% **efficient**
- ◦ Production Efficiency
- Trophic Efficiency and Ecological Pyramids
- **CONCEPT 55.4 Biological and geochemical processes cycle nutrients and water in ecosystems**
- ◦ Decomposition and Nutrient Cycling Rates
- Biogeochemical Cycles
- *Case Study:* Nutrient Cycling in the Hubbard Brook Experimental Forest
- **CONCEPT 55.5 Restoration ecologists return degraded ecosystems to a more natural state**
- ◦ Bioremediation
- Biological Augmentation
- Ecosystems: *A Review*

## 56 Conservation Biology and Global Change

- **CONCEPT 56.1 Human activities threaten earth's biodiversity**
- ◦ Three Levels of Biodiversity

- Biodiversity and Human Welfare
- Threats to Biodiversity
- Can Extinct Species Be Resurrected?
- **CONCEPT 56.2 Population conservation focuses on population size, genetic diversity, and critical habitat**
- ◦ Extinction Risks in Small Populations
- Critical Habitat
- Weighing Conflicting Demands
- **CONCEPT 56.3 Landscape and regional conservation help sustain biodiversity**
- ◦ Landscape Structure and Biodiversity
- Establishing Protected Areas
- Urban Ecology
- **CONCEPT 56.4 Earth is changing rapidly as a result of human actions**
- ◦ Nutrient Enrichment
- Toxins in the Environment
- Greenhouse Gases and Climate Change
- Depletion of Atmospheric Ozone
- **CONCEPT 56.5 Sustainable development can improve human lives while conserving biodiversity**
- ◦ Sustainable Development
- The Future of the Biosphere

# APPENDIX A

- **Answers**

# APPENDIX B

- **Classification of Life**

## APPENDIX C

- **A Comparison of the Light Microscope and the Electron Microscope**

## APPENDIX D

- **Scientific Skills Review**

## CREDITS

## GLOSSARY

## THE METRIC SYSTEM

## THE PERIODIC TABLE OF THE ELEMENTS

## AP® BIOLOGY EQUATIONS AND FORMULAS

# Skills Exercises

## Scientific Skills Exercises

- ****1Interpreting a Pair of Bar Graphs
- **2**Calibrating a Standard Radioactive Isotope Decay Curve and Interpreting Data
- ****3Interpreting a Scatter Plot with a Regression Line
- ****4Working with Moles and Molar Ratios

- ****5Analyzing Polypeptide Sequence Data
- ****6Using a Scale Bar to Calculate Volume and Surface Area of a Cell
- ****7Interpreting a Scatter Plot with Two Sets of Data
- ****8Making a Line Graph and Calculating a Slope
- ****9Making a Bar Graph and Evaluating a Hypothesis
- ****10Making Scatter Plots with Regression Lines
- ****11*Using Experiments to Test a Model*
- ****12Interpreting Histograms
- ****13Making a Line Graph and Converting Between Units of Data
- ****14Making a Histogram and Analyzing a Distribution Pattern
- ****15Using the Chi-Square (χ2) Test
- ****16Working with Data in a Table
- ****17Interpreting a Sequence Logo
- ****18Analyzing DNA Deletion Experiments
- ****19Analyzing a Sequence-Based Phylogenetic Tree to Understand Viral Evolution
- ****20*Analyzing Quantitative and Spatial Gene Expression Data*
- ****21Reading an Amino Acid Sequence Identity Table
- ****22Making and Testing Predictions
- ****23Using the Hardy-Weinberg Equation to Interpret Data and Make Predictions
- ****24Identifying Independent and Dependent Variables, Making a Scatter Plot, and Interpreting Data
- ****25Estimating Quantitative Data from a Graph and Developing Hypotheses
- ****26Using Protein Sequence Data to Test an Evolutionary Hypothesis
- ****27Calculating and Interpreting Means and Standard Errors
- Making a Bar Graph and Interpreting the Data*
- ****28Interpreting Comparisons of Genetic Sequences

- ****29Making Bar Graphs and Interpreting Data
- ****30Using Natural Logarithms to Interpret Data
- ****31Interpreting Genomic Data and Generating Hypotheses
- Synthesizing Information from Multiple Data Sets*
- 32Calculating and Interpreting Correlation Coefficients
- ****33Understanding Experimental Design and Interpreting Data
- ****34Determining the Equation of a Regression Line
- ****35Using Bar Graphs to Interpret Data
- ****36Calculating and Interpreting Temperature Coefficients
- ****37Making Observations
- ****38Using Positive and Negative Correlations to Interpret Data
- ****39Interpreting Experimental Results from a Bar Graph
- ****40Interpreting Pie Charts
- ****41Interpreting Data from an Experiment with Genetic Mutants
- ****42Making and Interpreting Histograms
- ****43Comparing Two Variables on a Common $x$-Axis
- ****44Describing and Interpreting Quantitative Data
- ****45Designing a Controlled Experiment
- ****46Making Inferences and Designing an Experiment
- ****47Interpreting a Change in Slope
- ****48Interpreting Data Values Expressed in Scientific Notation
- ****49Designing an Experiment using Genetic Mutants
- ****50Interpreting a Graph with Log Scales
- ****51Testing a Hypothesis with a Quantitative Model
- ****52Making a Bar Graph and a Line Graph to Interpret Data
- ****53Using the Logistic Equation to Model Population Growth

- ****54Making a Bar Graph and a Scatter Plot
- ****55Interpreting Quantitative Data
- ****56Graphing Data and Evaluating Evidence

*Available only in Mastering Biology. All other Scientific Skills Exercises are in the print book, eTextbook, and Mastering Biology.

## Problem-Solving Exercises

- ****5Are you a victim of fish fraud?
- ****11Can a skin wound turn deadly?
- ****17Are insulin mutations the cause of three infants' neonatal diabetes?
- ****24Is hybridization promoting insecticide resistance in mosquitoes that transmit malaria?
- ****34Can declining amphibian populations be saved by a vaccine?
- ****39How will climate change affect crop productivity?
- ****45Is thyroid regulation normal in this patient?
- ****55Can an insect outbreak threaten a forest's ability to absorb $CO_2$ from the atmosphere?

---

# Featured Figures

## Visualizing Figures

- ****5.16Visualizing Proteins

Figure 5.16 Visualizing proteins. Structural models of proteins: space-filling model, ribbon model, and wireframe model are shown. Simplified diagrams of proteins in terms of functions: transparent shape, solid shape, simple shape, and as a dot.

- ****6.32Visualizing the Scale of the Molecular Machinery in a Cell
- ****16.7Visualizing DNA
- ****25.8Visualizing the Scale of Geologic Time

- ****26.5Visualizing Phylogenetic Relationships
- ****32.8Visualizing Animal Body Symmetry and Axes
- ****35.11Visualizing Primary and Secondary Growth
- ****47.8Visualizing Gastrulation
- ****55.13Visualizing Biogeochemical Cycles

## Make Connections Figures

- ****5.26Contributions of Genomics and Proteomics to Biology
- ****10.22The Working Cell
- ****18.27Genomics, Cell Signaling, and Cancer
- ****23.18The Sickle-Cell Allele
- ****33.8Maximizing Surface Area
- ****37.9Mutualism Across Kingdoms and Domains
- ****39.27Levels of Plant Defenses Against Herbivores
- ****40.23Life Challenges and Solutions in Plants and Animals
- ****44.17Ion Movement and Gradients
- ****55.19The Working Ecosystem
- ****56.31Climate Change Has Effects at All Levels of Biological Organization

## Exploring Figures

- ****1.3Levels of Biological Organization
- ****5.18Levels of Protein Structure
- ****6.3Microscopy
- ****6.8Eukaryotic Cells
- ****6.30Cell Junctions in Animal Tissues
- ****7.21Endocytosis in Animal Cells

- ****11.8Cell-Surface Transmembrane Receptors
- ****12.7Mitosis in an Animal Cell
- ****13.8Meiosis in an Animal Cell
- ****16.23Chromatin Packing in a Eukaryotic Chromosome
- ****24.3Reproductive Barriers
- ****25.7The Origin of Mammals
- ****27.17Bacterial Diversity
- ****28.5Protistan Diversity
- ****29.5Alternation of Generations
- ****29.13Bryophyte Diversity
- ****29.19Seedless Vascular Plant Diversity
- ****30.7Gymnosperm Diversity
- ****30.17Angiosperm Diversity
- ****33.42Insect Diversity
- ****34.42Mammalian Diversity
- ****35.10Examples of Differentiated Plant Cells
- ****41.5Four Main Feeding Mechanisms of Animals
- ****44.12The Mammalian Excretory System
- ****46.11Human Gametogenesis
- ****49.11The Organization of the Human Brain
- ****50.10The Structure of the Human Ear
- ****50.17The Structure of the Human Eye
- ****53.17Mechanisms of Density-Dependent Regulation
- ****55.14Water and Nutrient Cycling
- ****55.17Restoration Ecology Worldwide

# Inquiry Figures

- ****1.25Does camouflage affect predation rates on two populations of mice?
- ****4.2Can organic molecules form under conditions estimated to simulate those on the early Earth?
- ****7.4Do membrane proteins move?
- †10.9Which wavelengths of light are most effective in driving photosynthesis?
- ****12.9At which end do kinetochore microtubules shorten during anaphase?
- ****12.14Do molecular signals in the cytoplasm regulate the cell cycle?
- ****14.3When F1 hybrid pea plants self- or cross-pollinate, which traits appear in the F2 generation?
- ****14.8Do the alleles for one character segregate into gametes dependently or independently of the alleles for a different character?
- †15.3In a cross between a wild-type female fruit fly and a mutant white-eyed male, what color eyes will the F1 and F2 offspring have?
- ****15.9How does linkage between two genes affect inheritance of characters?
- ****16.2Can a genetic trait be transferred between different bacterial strains?
- ****16.4Is protein or DNA the genetic material of phage T2?
- *†16.12Does DNA replication follow the conservative, semiconservative, or dispersive model?
- ****17.3Do individual genes specify the enzymes that function in a biochemical pathway?
- ****18.22Could Bicoid be a morphogen that determines the anterior end of a fruit fly?
- ****19.2What causes tobacco mosaic disease?
- ****20.16Can the nucleus from a differentiated animal cell direct development of an organism?
- ****20.21Can a fully differentiated human cell be “deprogrammed” to become a stem cell?
- ****21.18What is the function of a gene (*FOXP2*) that may be involved in language acquisition?
- ****22.13Can a change in a population’s food source result in evolution by natural selection?
- *23.16Do females select mates based on traits indicative of “good genes”?

- ****24.7Can divergence of allopatric populations lead to reproductive isolation?
- ****24.12Does sexual selection in cichlids result in reproductive isolation?
- ****24.19How does hybridization lead to speciation in sunflowers?
- ****25.27What causes the loss of spines in lake stickleback fish?
- ****26.6What is the species identity of food being sold as whale meat?
- ****28.26What is the root of the eukaryotic tree?
- ****29.14Can bryophytes reduce the rate at which key nutrients are lost from soils?
- ****31.22Do fungal endophytes benefit a woody plant?
- ****33.29Did the arthropod body plan result from new *Hox* genes?
- ****36.18Does phloem sap contain more sugar near sources than near sinks?
- ****37.10How variable are the compositions of bacterial communities inside and outside of roots?
- ****39.5What part of a grass coleoptile senses light, and how is the signal transmitted?
- ****39.6What causes polar movement of auxin from shoot tip to base?
- ****39.16How does the order of red and far-red illumination affect seed germination?
- ****40.16How does a Burmese python generate heat while incubating eggs?
- ****40.22What happens to the circadian clock during hibernation?
- *41.4Can diet influence the frequency of neural tube defects?
- ****42.25What causes respiratory distress syndrome?
- ****44.20Can aquaporin mutations cause diabetes?
- ****46.8Why is sperm usage biased when female fruit flies mate twice?
- †47.3Does the distribution of Ca2+ in an egg correlate with formation of the fertilization envelope?
- ****47.23How does distribution of the gray crescent affect the developmental potential of the first two daughter cells?
- ****47.24Can the dorsal lip of the blastopore induce cells in another part of the amphibian embryo to change their developmental fate?

- ****47.26What role does the zone of polarizing activity (ZPA) play in limb pattern formation in vertebrates?
- ****51.8Does a digger wasp use landmarks to find her nest?
- ****51.24Are differences in migratory orientation within a species genetically determined?
- ****53.13How does caring for offspring affect parental survival in kestrels?
- †54.3Can a species' niche be influenced by competition?
- ****54.20Is *Pisaster ochraceus* a keystone species?
- ****55.6Which nutrient limits phytoplankton production along the coast of Long Island?
- ****55.12How does temperature affect litter decomposition in an ecosystem?
- *56.12What caused the drastic decline of the Illinois greater prairie chicken population?

## Research Method Figures

- ****5.21X-Ray Crystallography
- ****6.4Cell Fractionation
- ****10.8Determining an Absorption Spectrum
- ****13.3Preparing a Karyotype
- ****14.2Crossing Pea Plants
- ****14.7The Testcross
- ****15.11Constructing a Linkage Map
- ****20.3Sequencing by Synthesis: Next-Generation Sequencing
- ****20.7The Polymerase Chain Reaction (PCR)
- ****20.11RT-PCR Analysis of the Expression of Single Genes
- ****26.15Applying Parsimony to a Problem in Molecular Systematics
- ****35.21Using Dendrochronology to Study Climate
- ****37.7Hydroponic Culture
- ****48.6Intracellular Recording
- ****53.2Determining Population Size Using the Mark-Recapture Method
- ****54.14Determining Microbial Diversity Using Molecular Tools

A related Experimental Inquiry Tutorial can be assigned in Mastering Biology.

A related Experimental Inquiry Tutorial can be assigned in Mastering Biology.

A related Experimental Inquiry Tutorial can be assigned in Mastering Biology.

The Inquiry Figure, original research paper, and a worksheet to guide you through the paper are provided in *Inquiry in Action: Interpreting Scientific Papers*, Fourth Edition.

A related Experimental Inquiry Tutorial can be assigned in Mastering Biology.

The Inquiry Figure, original research paper, and a worksheet to guide you through the paper are provided in *Inquiry in Action: Interpreting Scientific Papers*, Fourth Edition.

The Inquiry Figure, original research paper, and a worksheet to guide you through the paper are provided in *Inquiry in Action: Interpreting Scientific Papers*, Fourth Edition.

A related Experimental Inquiry Tutorial can be assigned in Mastering Biology.

A related Experimental Inquiry Tutorial can be assigned in Mastering Biology.

The Inquiry Figure, original research paper, and a worksheet to guide you through the paper are provided in *Inquiry in Action: Interpreting Scientific Papers*, Fourth Edition.

---